Nonclassicality, quasiprobabilities and weak values explored in neutron interferometry

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Overview

- Introduction
- Nonclassicality
- Quasiprobabilities and weak values
- Experimental results

Introduction

Motivation and goal

Quantum mechanics exhibits features with no classical counterpart.

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However, not all quantum mechanics is non-classical and identifying the classical/quantum boundary is of importance.

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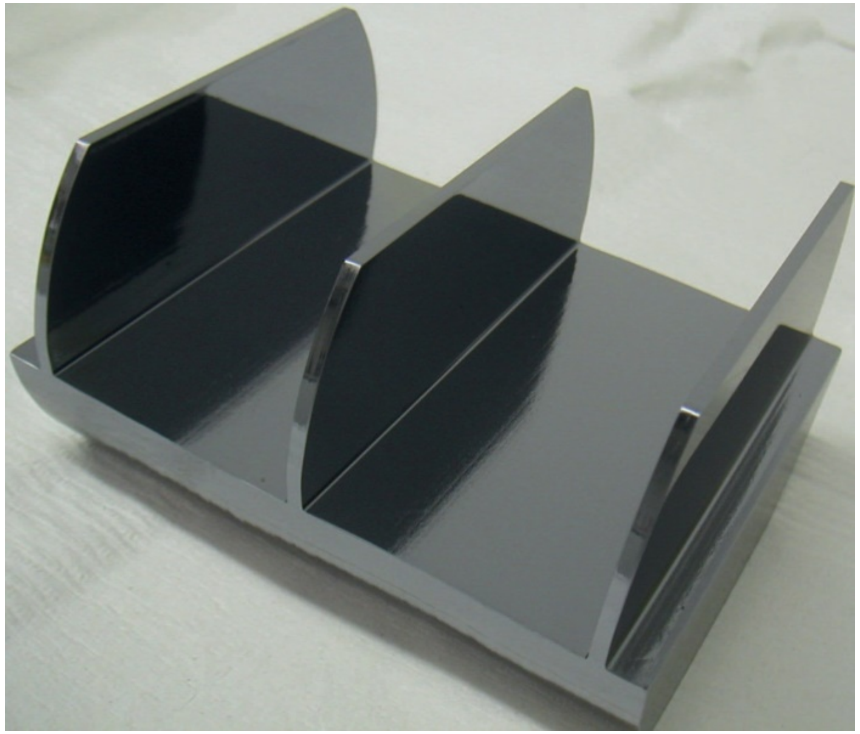
Quantum mechanics exhibits features with no classical counterpart.

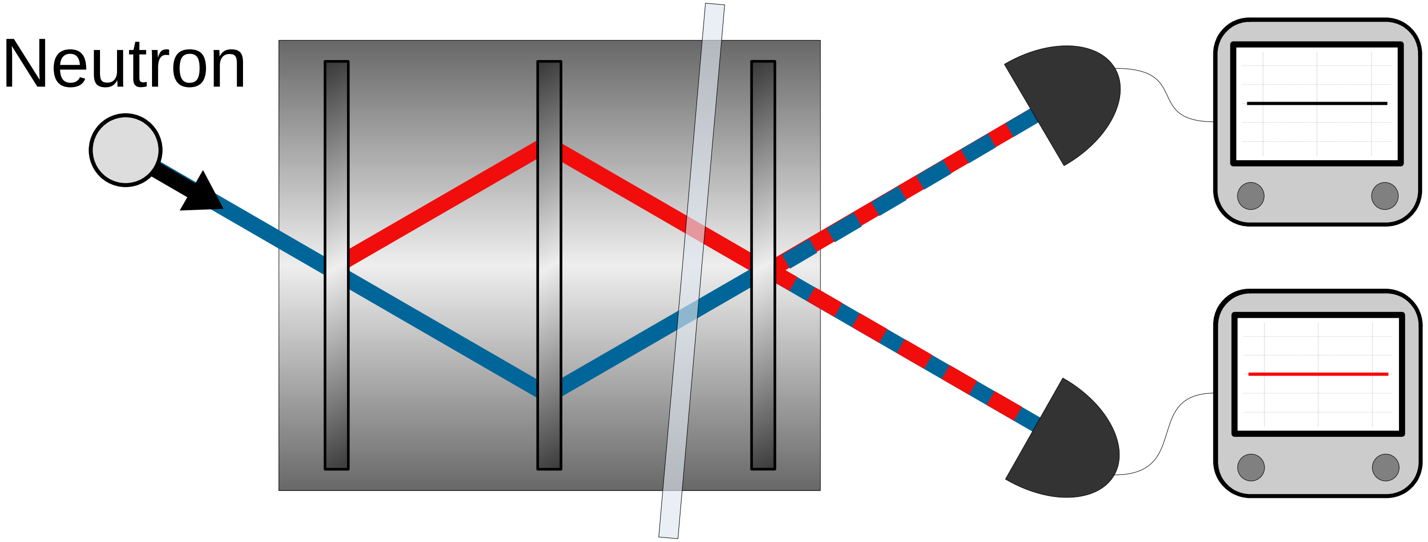
However, not all quantum mechanics is non-classical and identifying the classical/quantum boundary is of importance.

Quasiprobability representations and weak values provide a valuable tool to probe such boundaries.

Neutron interferometry

Neutron Mach-Zehnder interferometer:





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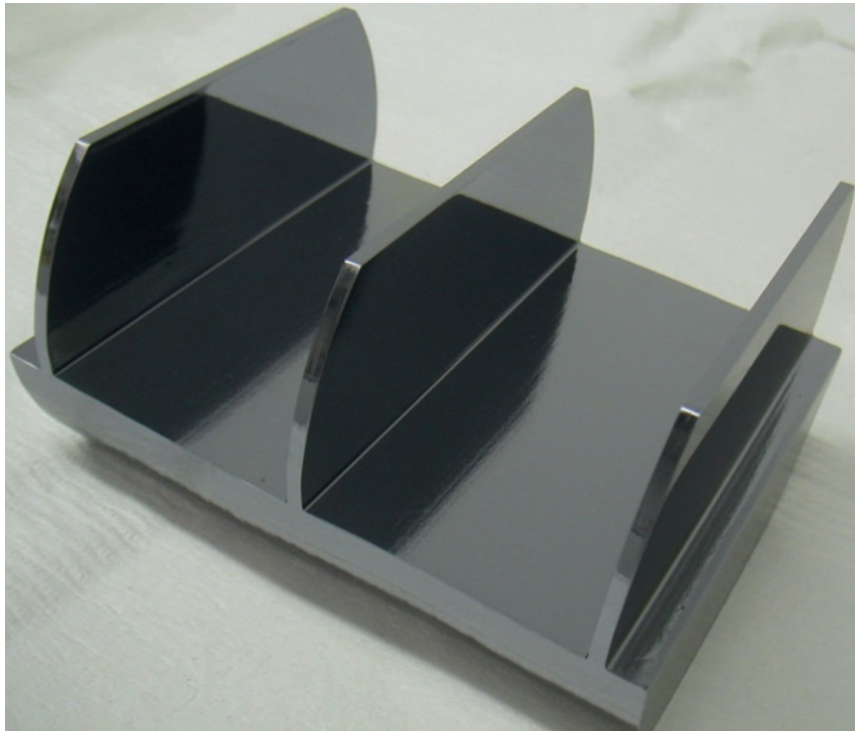
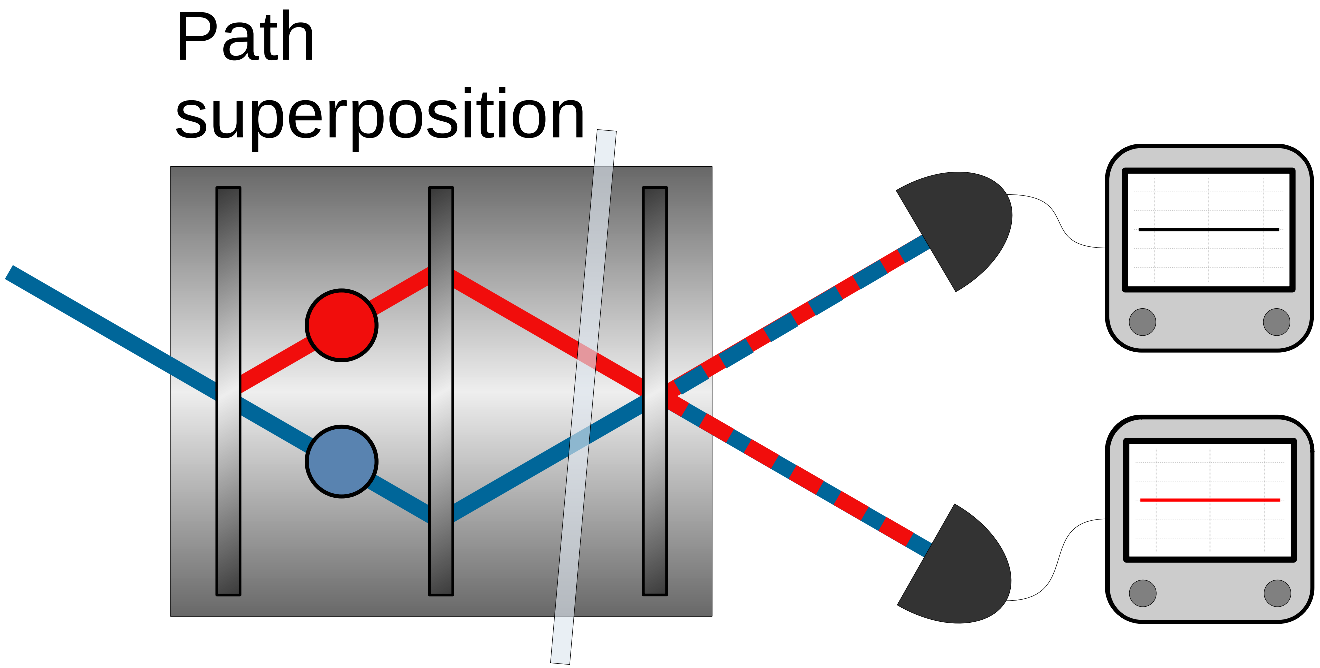
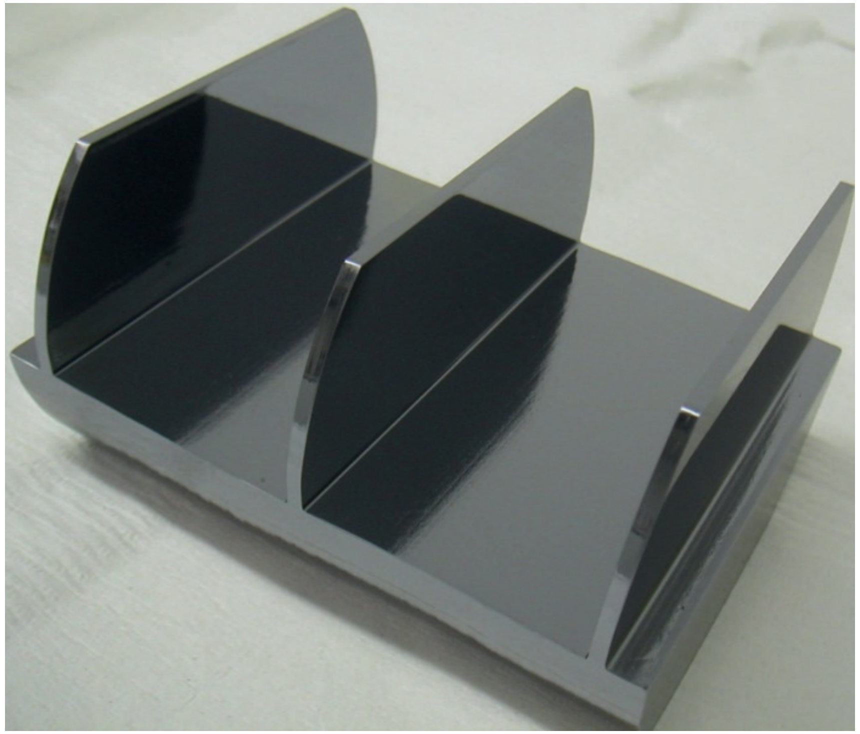
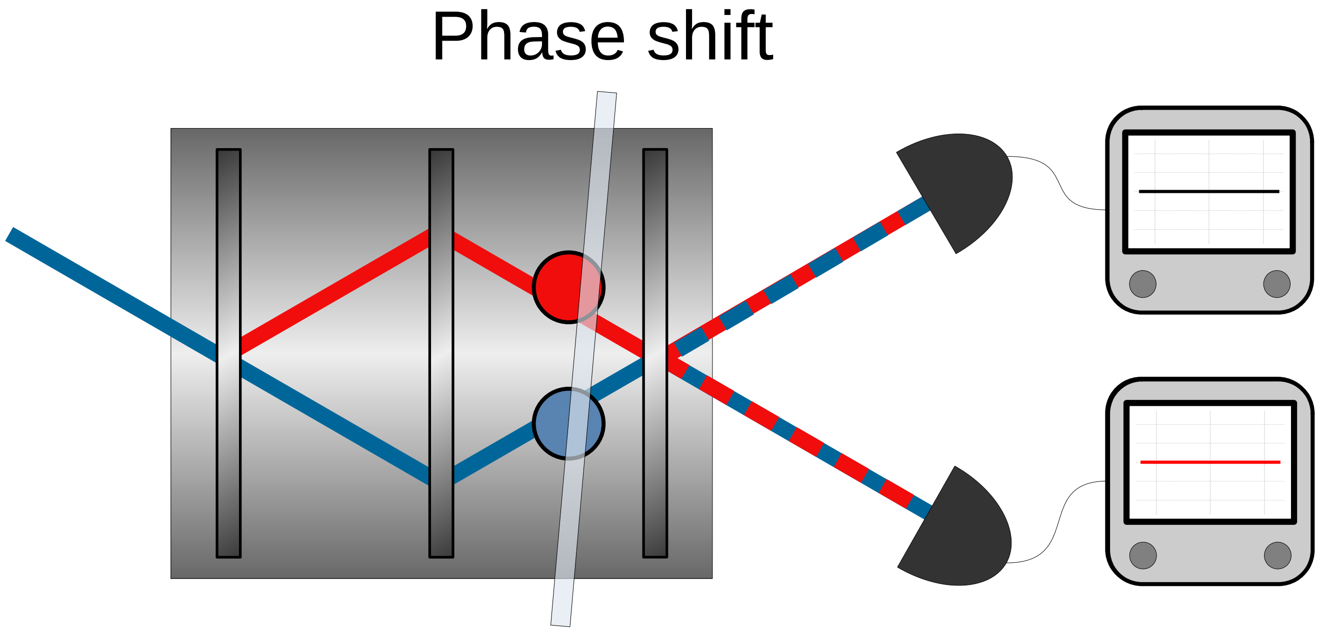




Neutron Mach-Zehnder interferometer:





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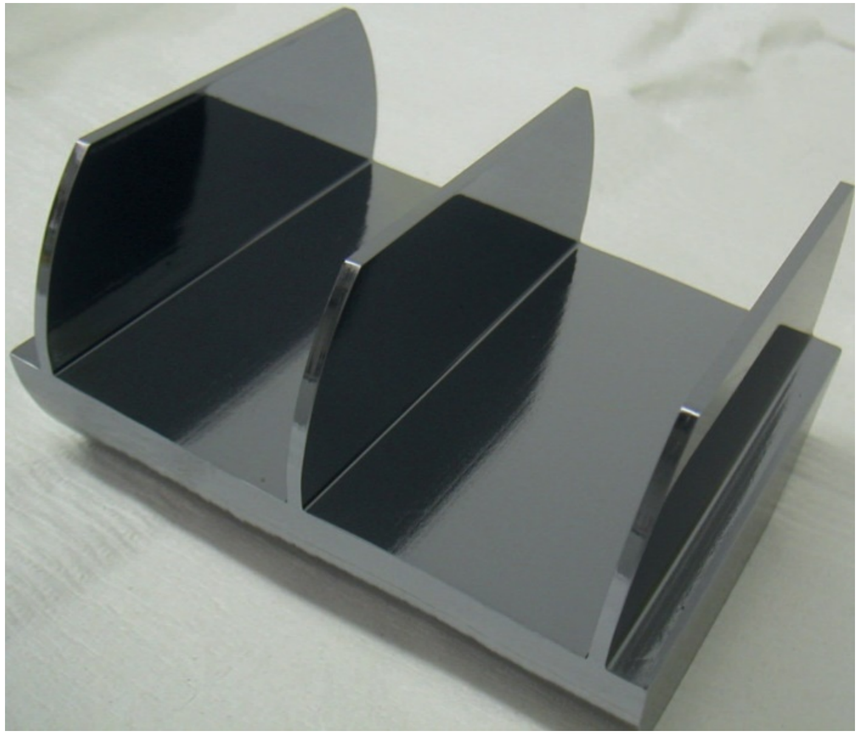
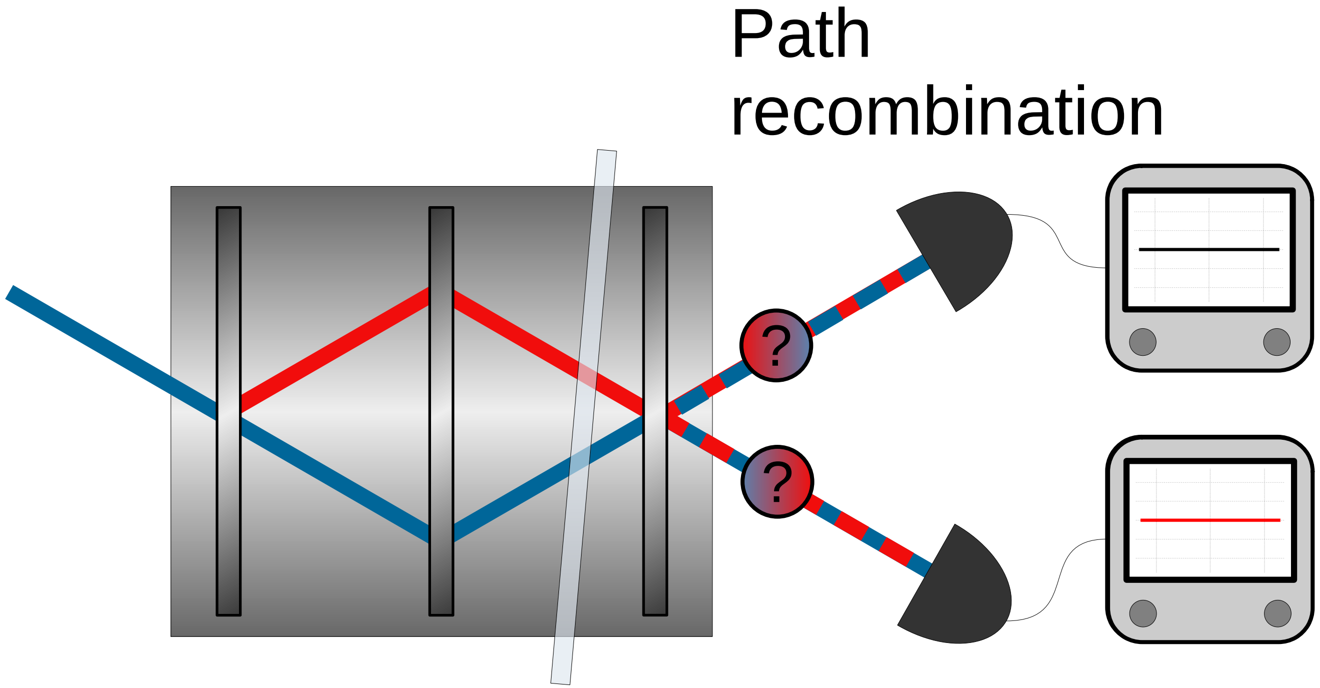
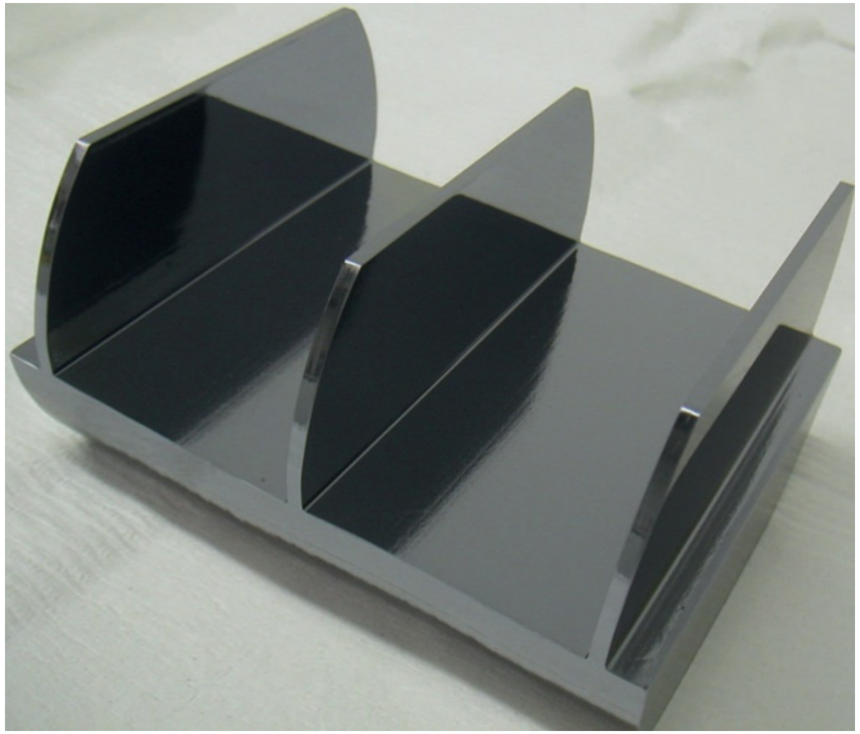
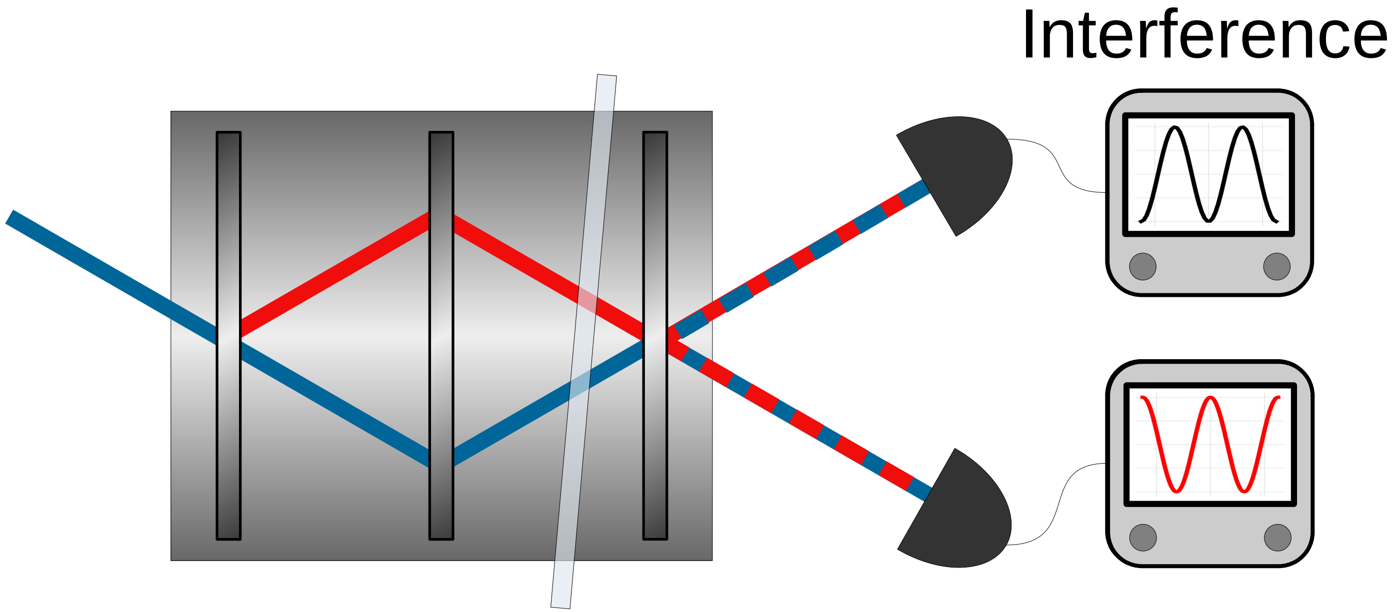




Neutron interferometry

Neutron Mach-Zehnder interferometer:





Why neutron interferometry?

- Interference is intrinsically a single-massive-particle phenomenon.
- Matter-wave interferometry at room temperature and ambient pressure, with macroscopic beam separation and high contrast.
- Manipulation of several degrees of freedom (path, spin, energy).

Sponar, S., Sedmik, R. I., Pitschmann, M., Abele, H. & Hasegawa, Y. Tests of fundamental quantum mechanics and dark interactions with low-energy neutrons. Nat. Rev. Phys. 3, 309–327 (2021).

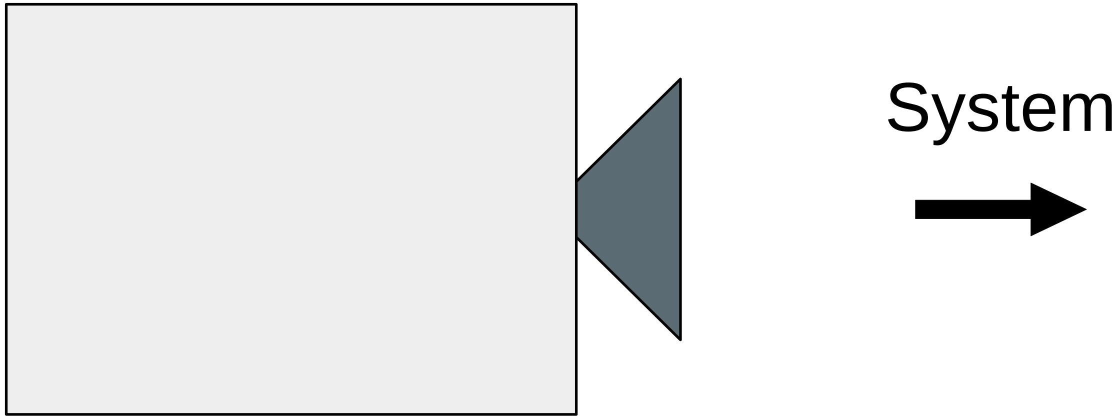
Nonclassicality

Preparation and Measurement

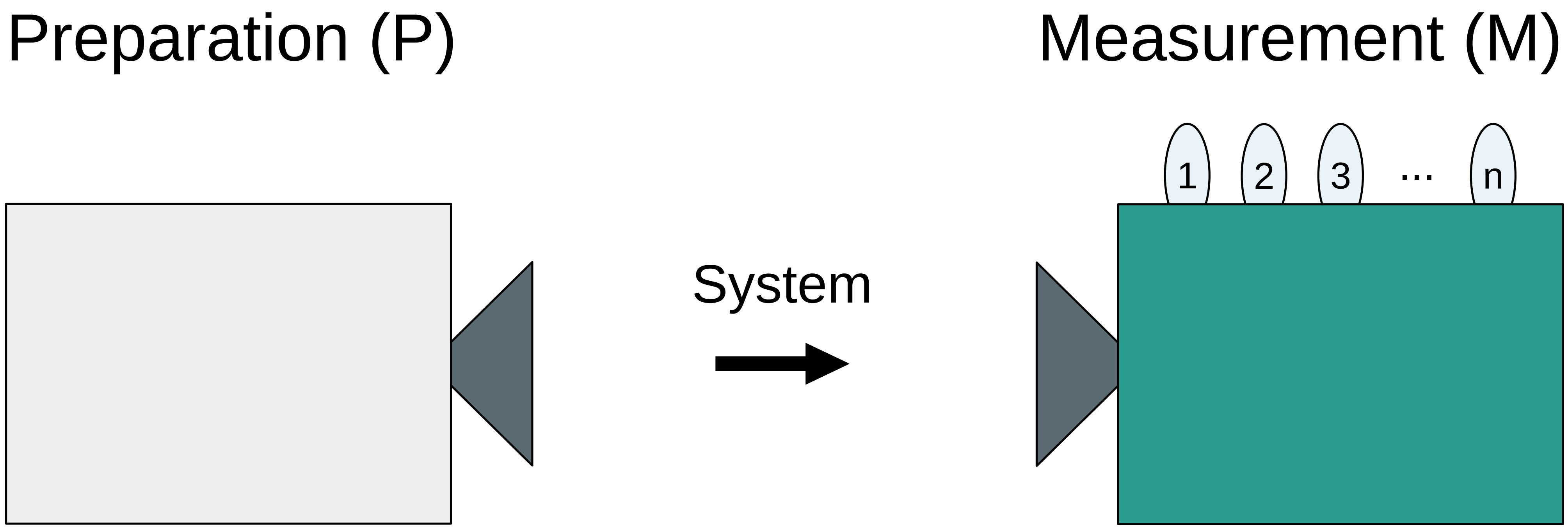
Preparation (P)

Preparation and Measurement

Preparation (P)



Preparation and Measurement



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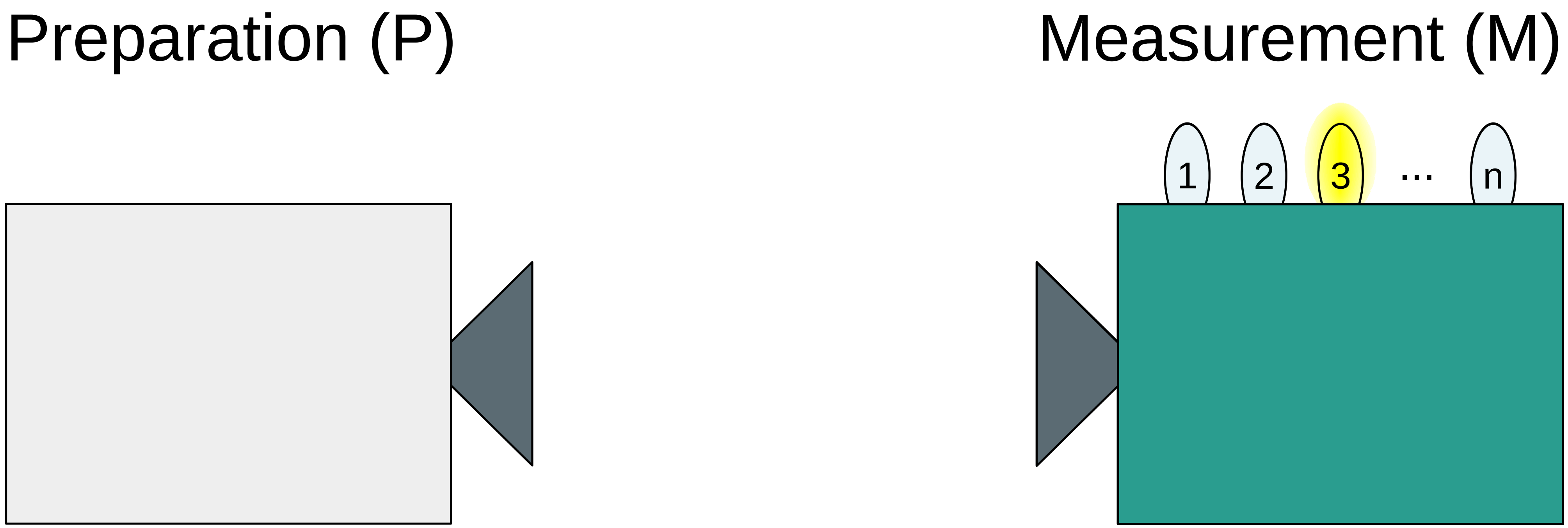

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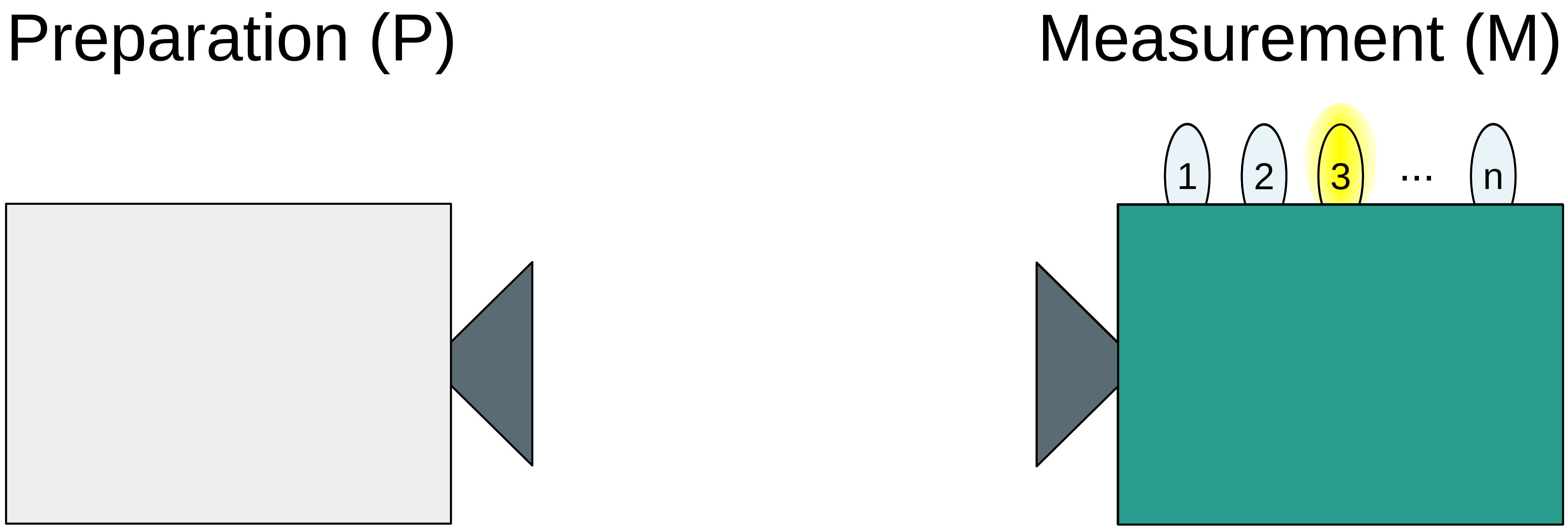
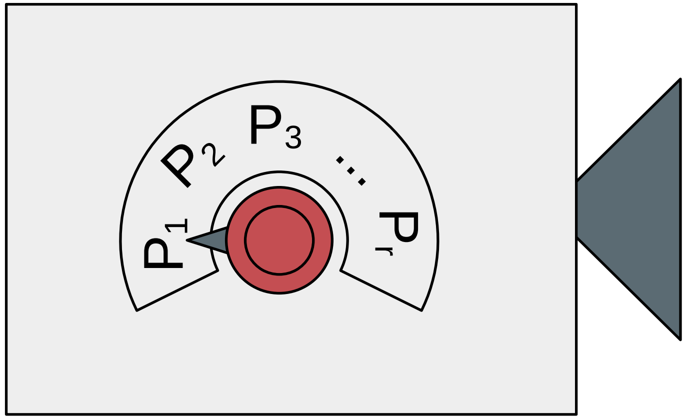


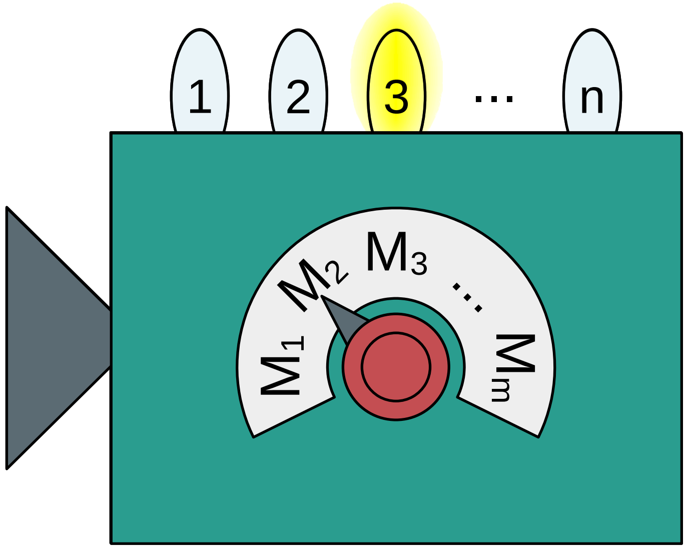
Outcome probabilities: $\mathrm{Prob}(k|P, M)$

Preparation and Measurement

Preparation (P_i)

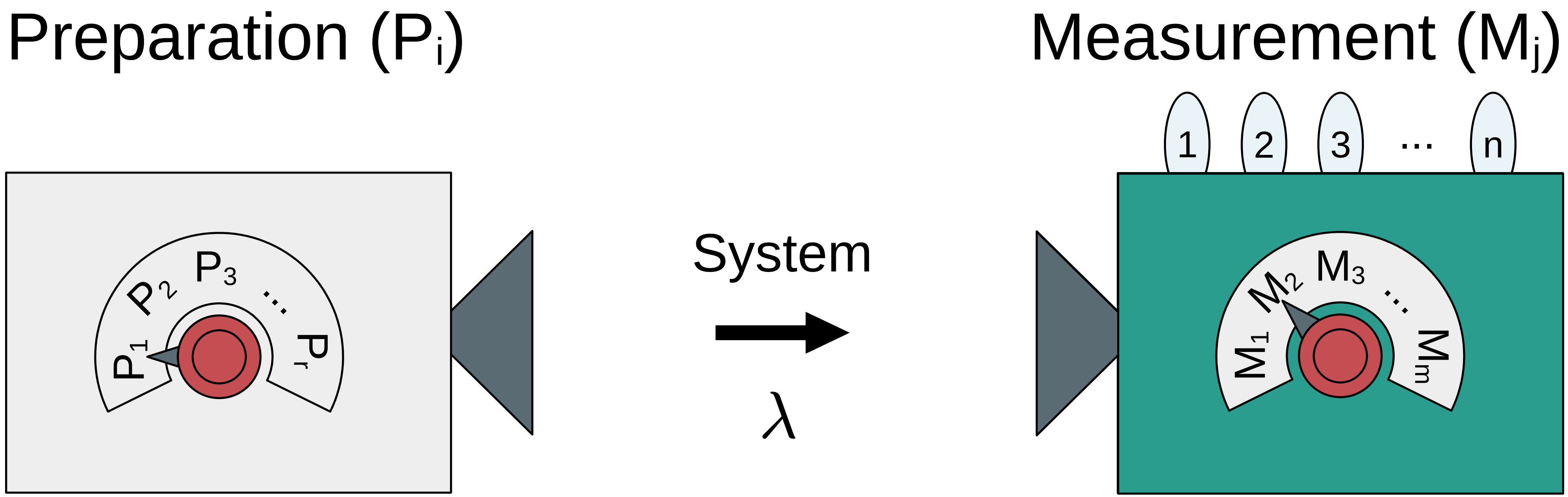


Measurement (M_j)



Outcome probabilities: $\mathrm{Prob}(k|P_i, M_j)$

A classical experiment



Defined set of properties: $\lambda = (\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_m)$

A classical experiment

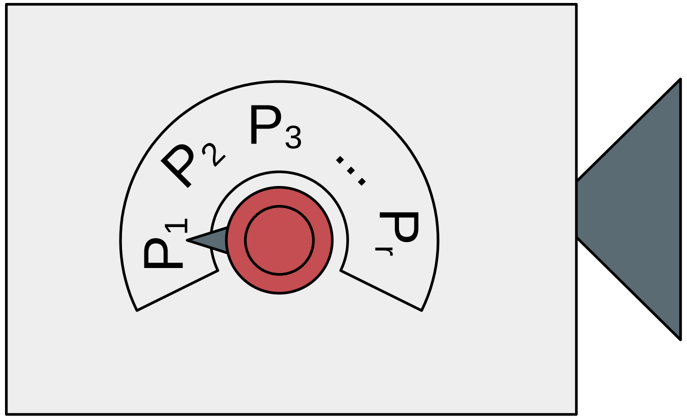
Preparation (P_i) Measurement (M_j)

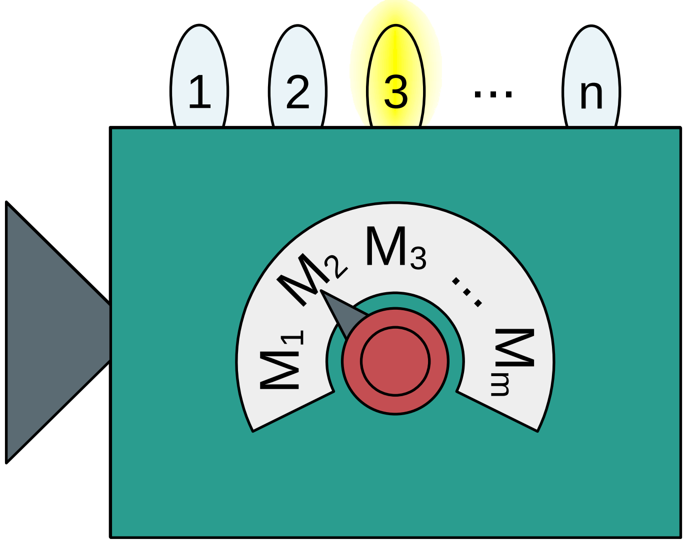
Defined set of properties: $\lambda = (\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_m)$

A classical experiment

Preparation (P_i)

Measurement (M_j)





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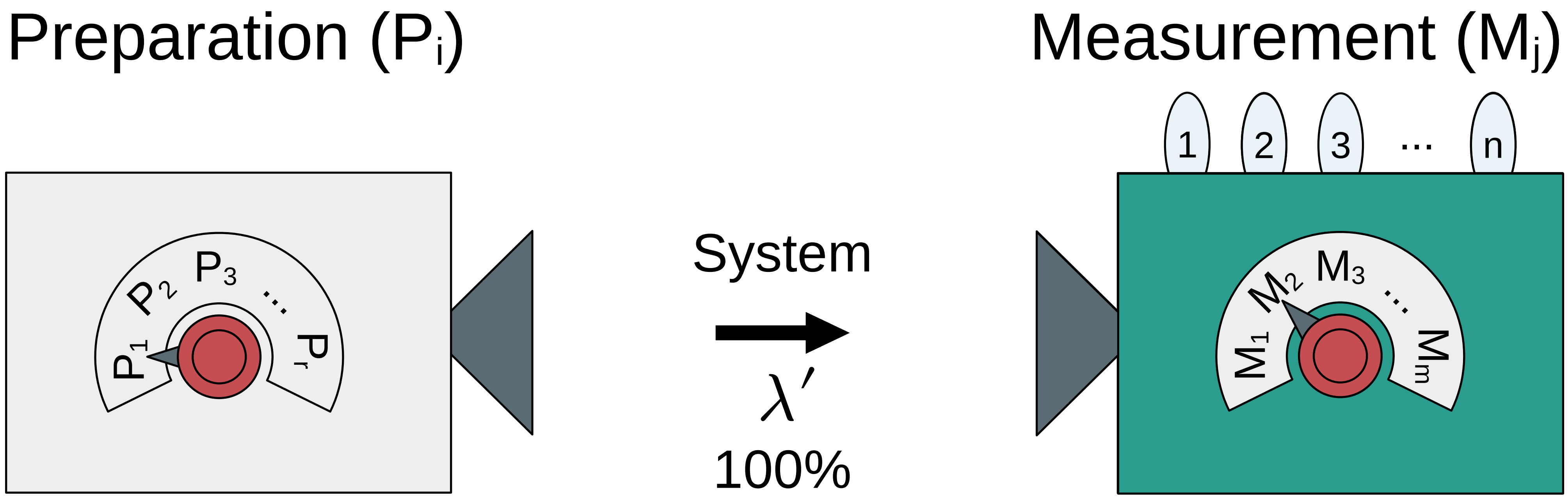
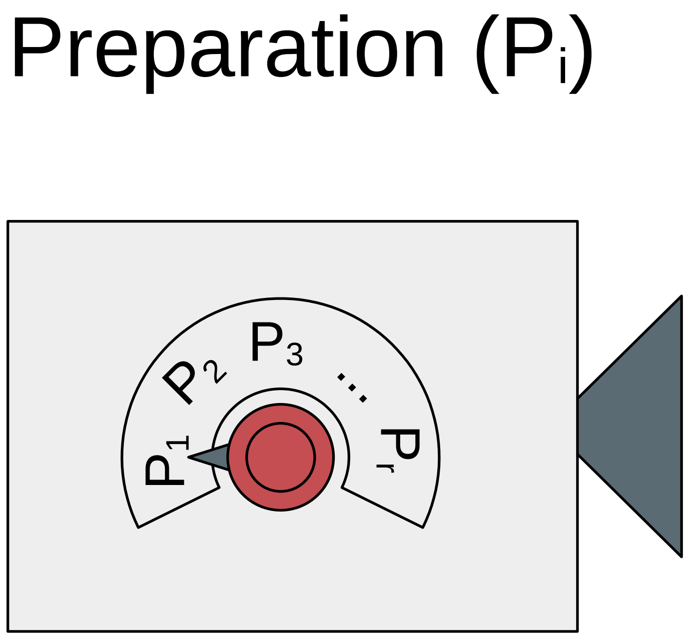
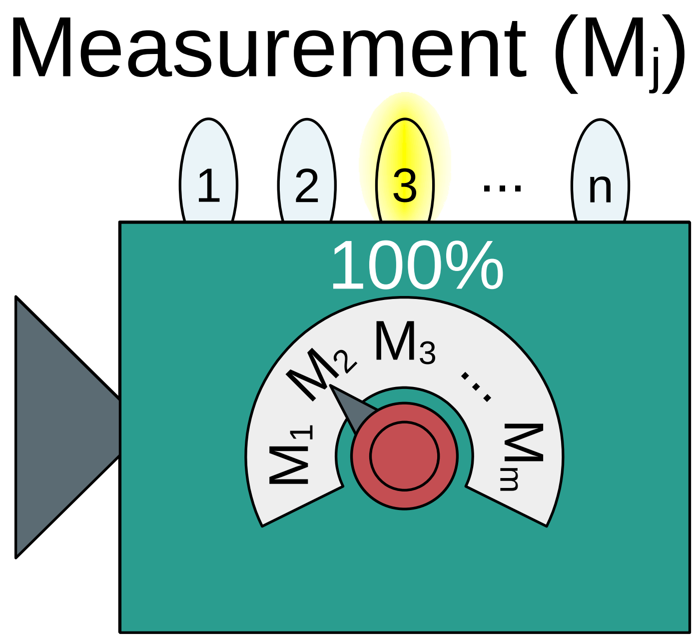


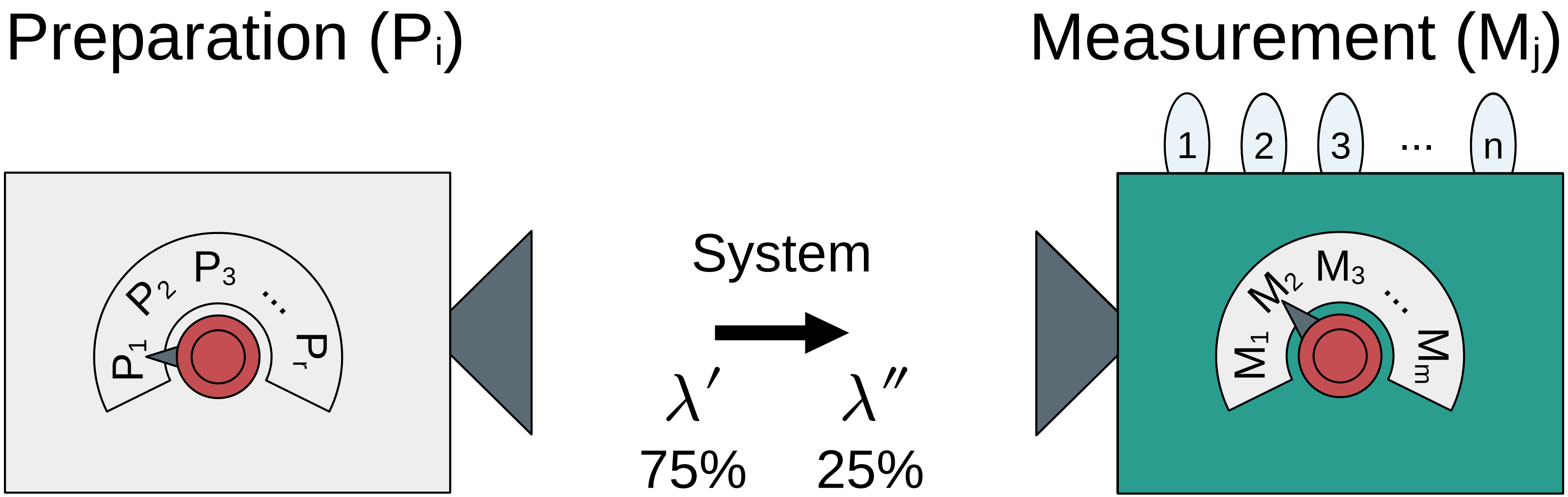
Preparation (P_i)
P_1
P_2
P_3
...
P_r
System
λ'
100%
Measurement (M_j)
1
2
3
...
n
M_1
M_2
M_3
...
M_m

A classical experiment



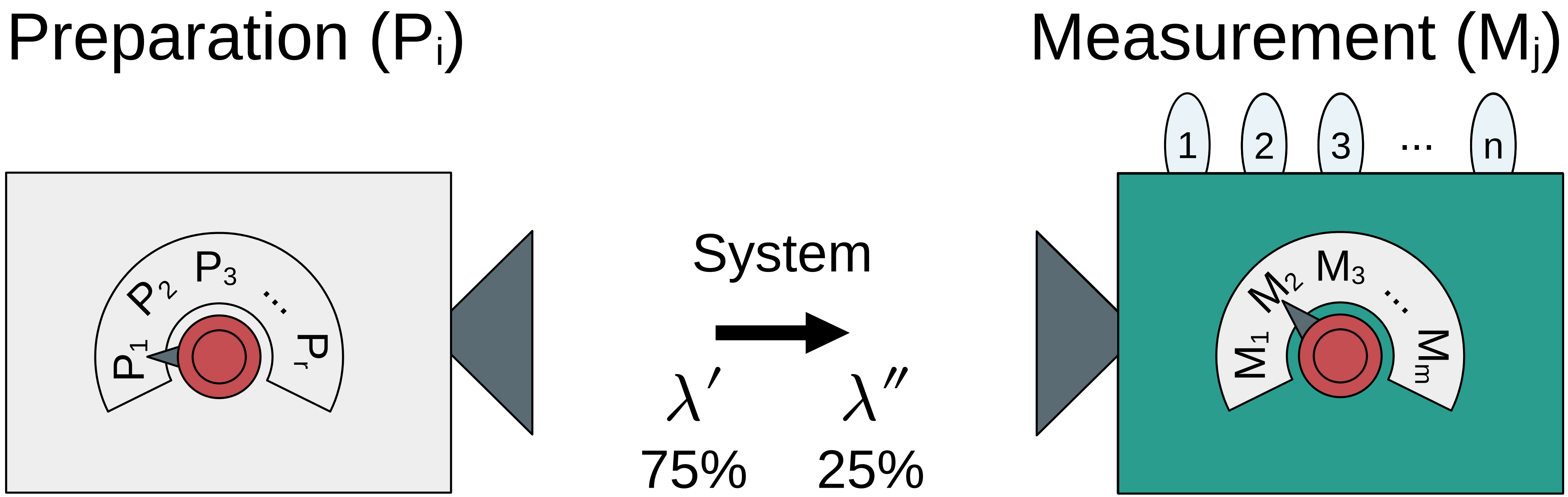


A classical experiment

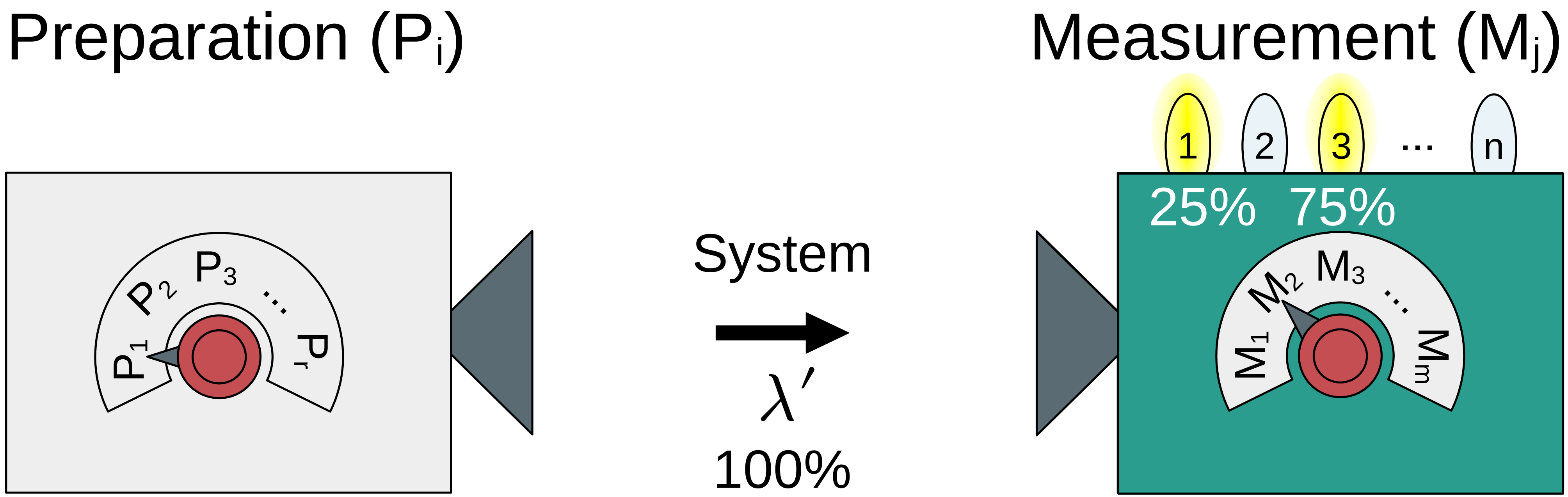


A classical experiment

Probability distribution: $\mu(\lambda|P_i)$

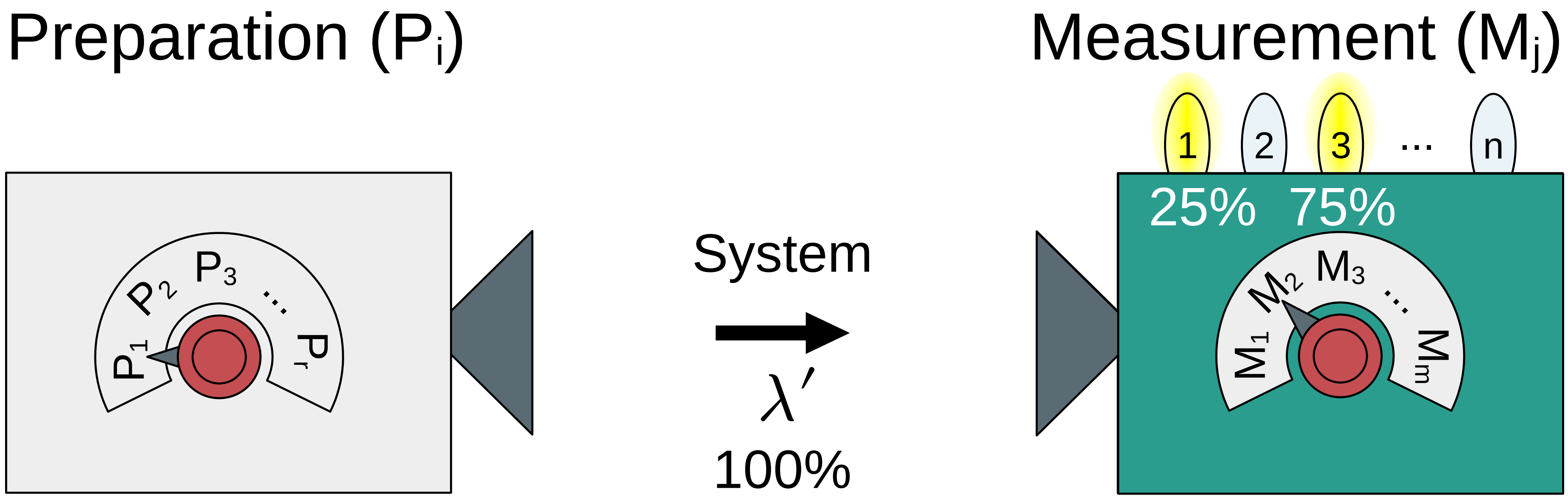


Probability distribution: $\mu(\lambda|P_i)$



A classical experiment

Probability distribution: $\mu(\lambda|P_i)$ Response function: $\xi(k|M_j, \lambda)$



A classical experiment

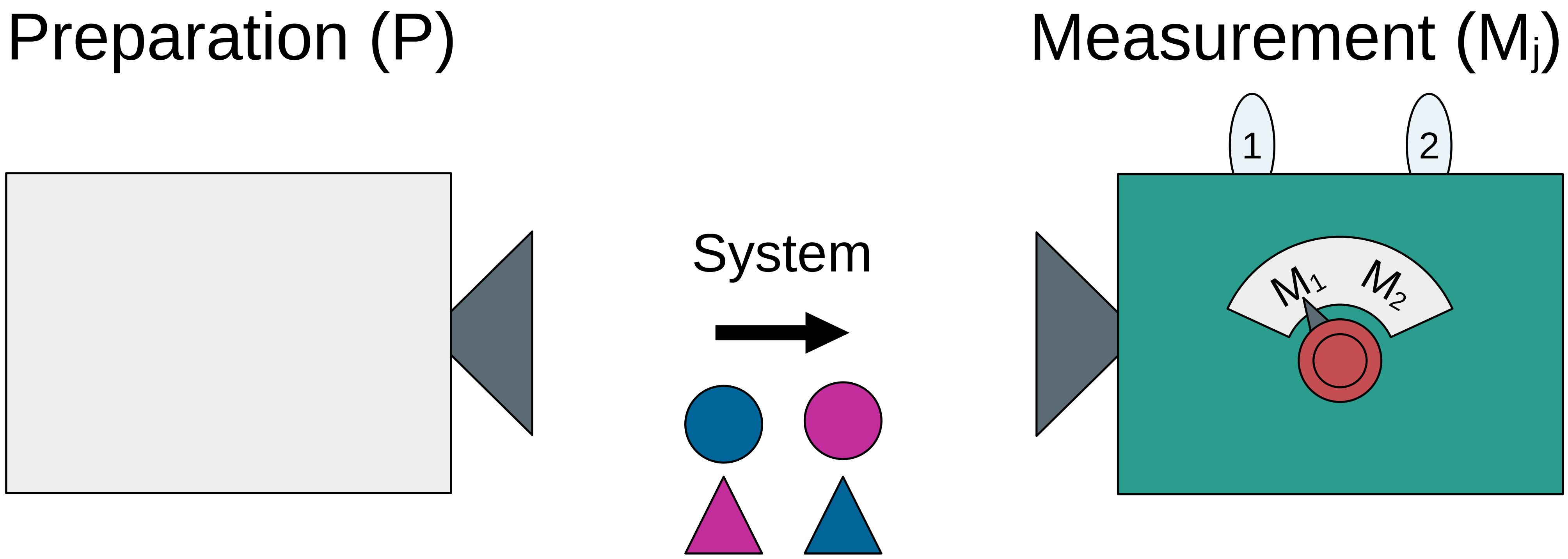
Probability distribution: $\mu(\lambda|P_i)$

Response function: $\xi(k|M_j, \lambda)$

Outcome probabilities:

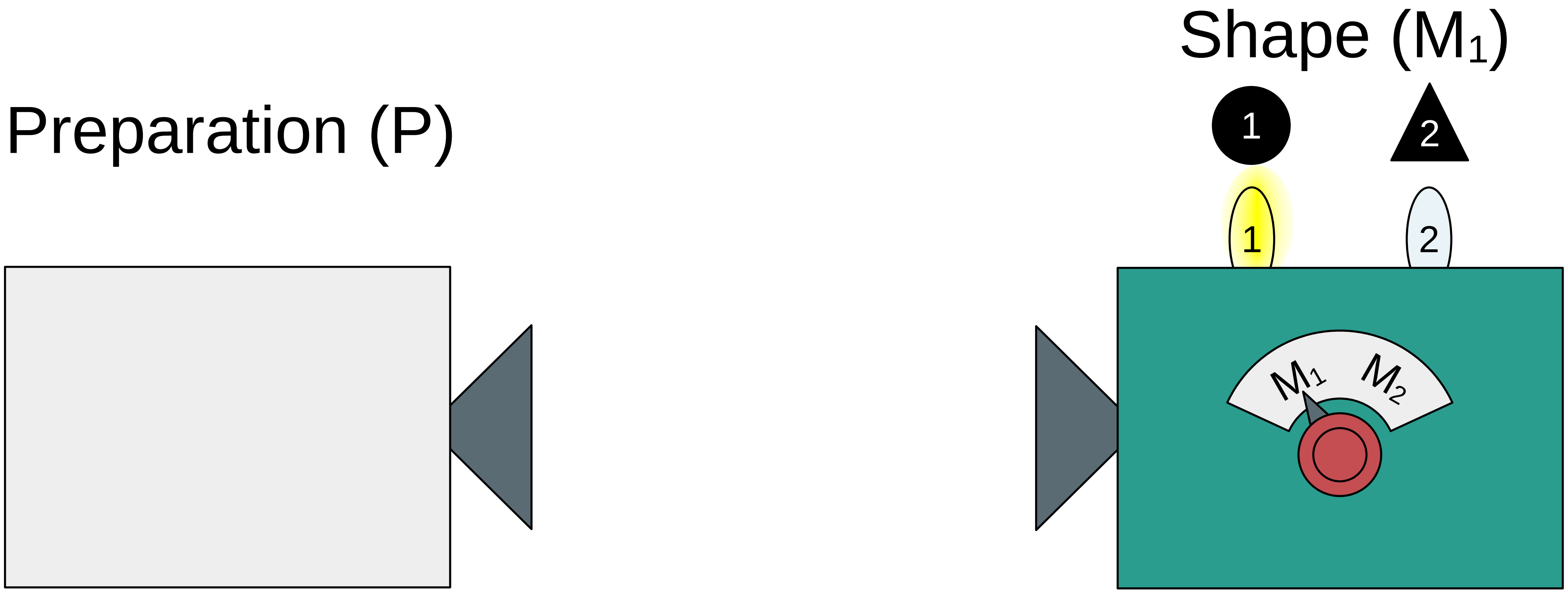
$$\mathrm{Prob}(k|P_i, M_j) = \sum_{\lambda} \mu(\lambda|P_i)\xi(k|M_j, \lambda)$$

A classical experiment



Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

A classical experiment



Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

A classical experiment

Color (M_1)

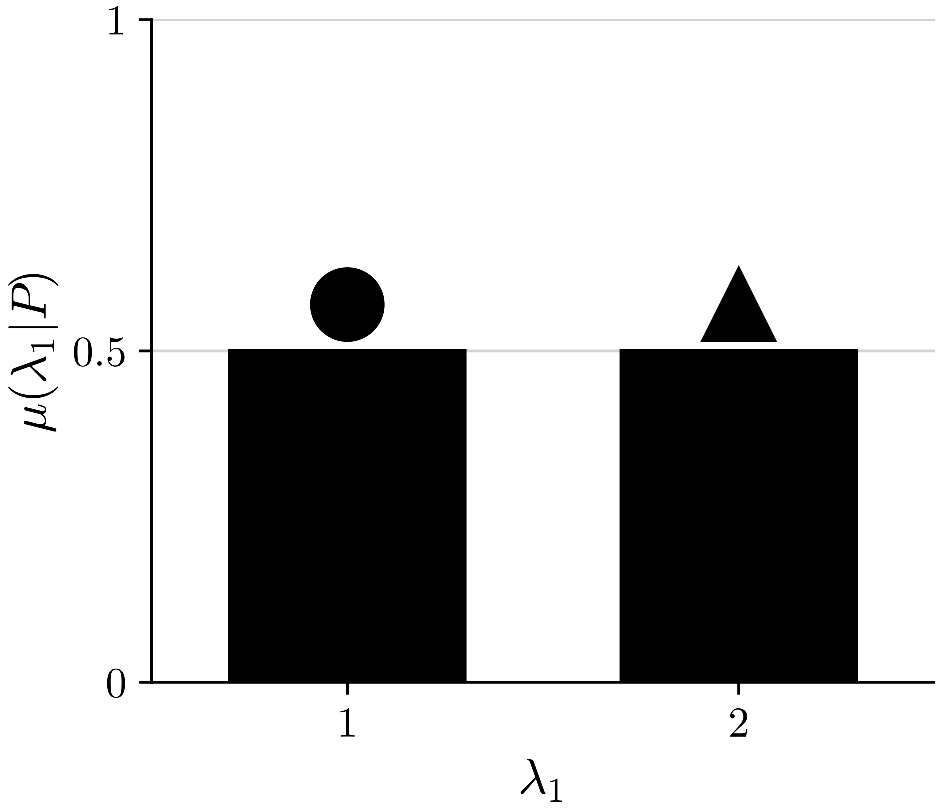
Preparation (P)

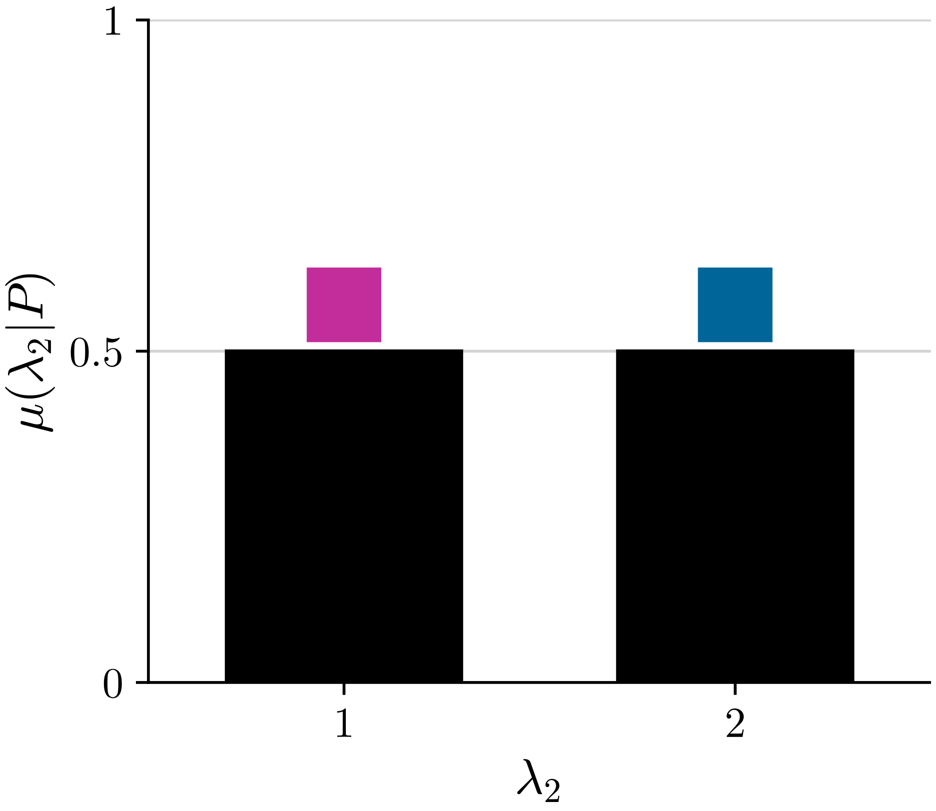
Two properties (shape, color): $\lambda = (\lambda_1, \lambda_2)$

A classical experiment

Outcome probabilities: ▲ 50% ● 50%

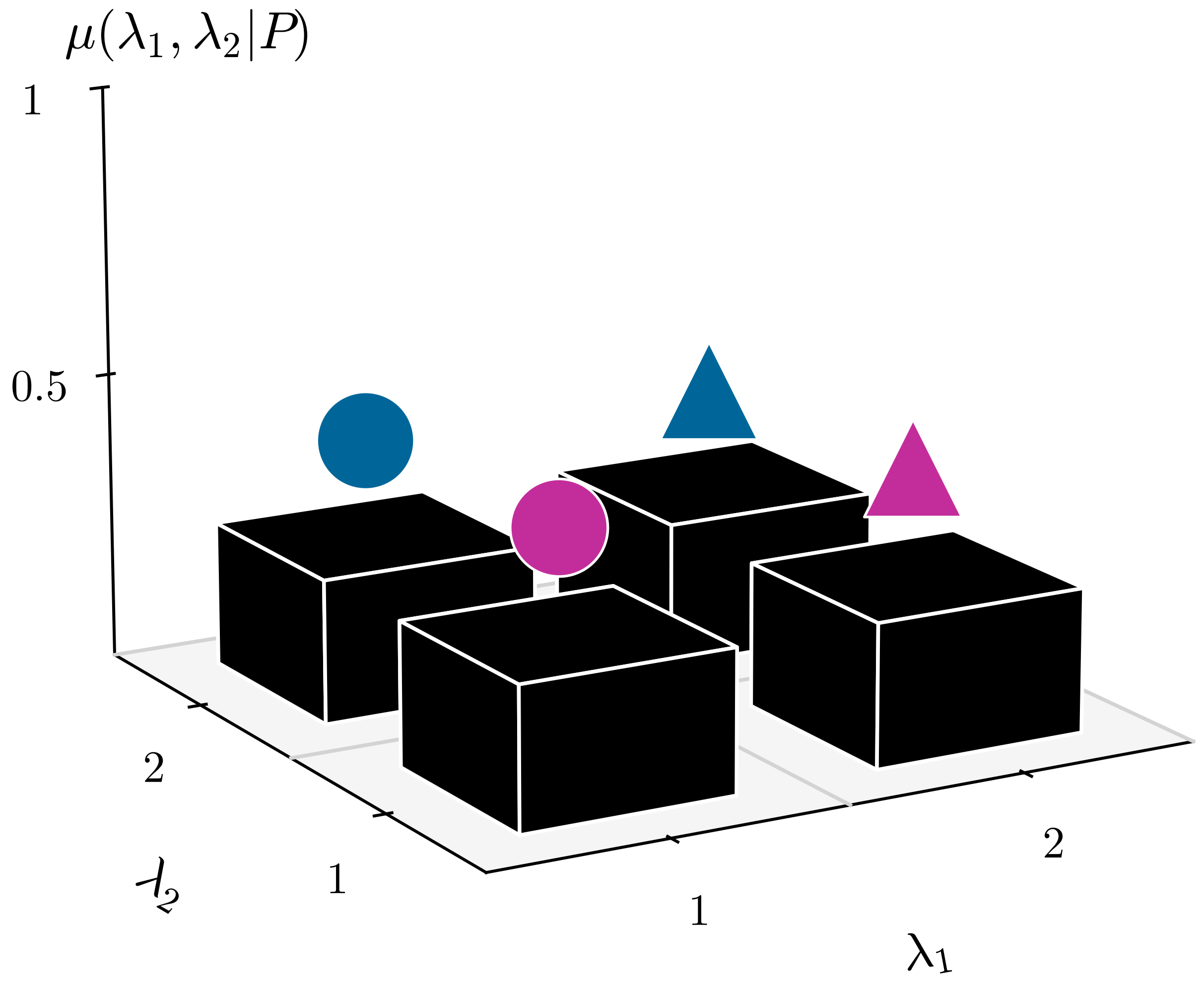
■ 50% ■ 50%

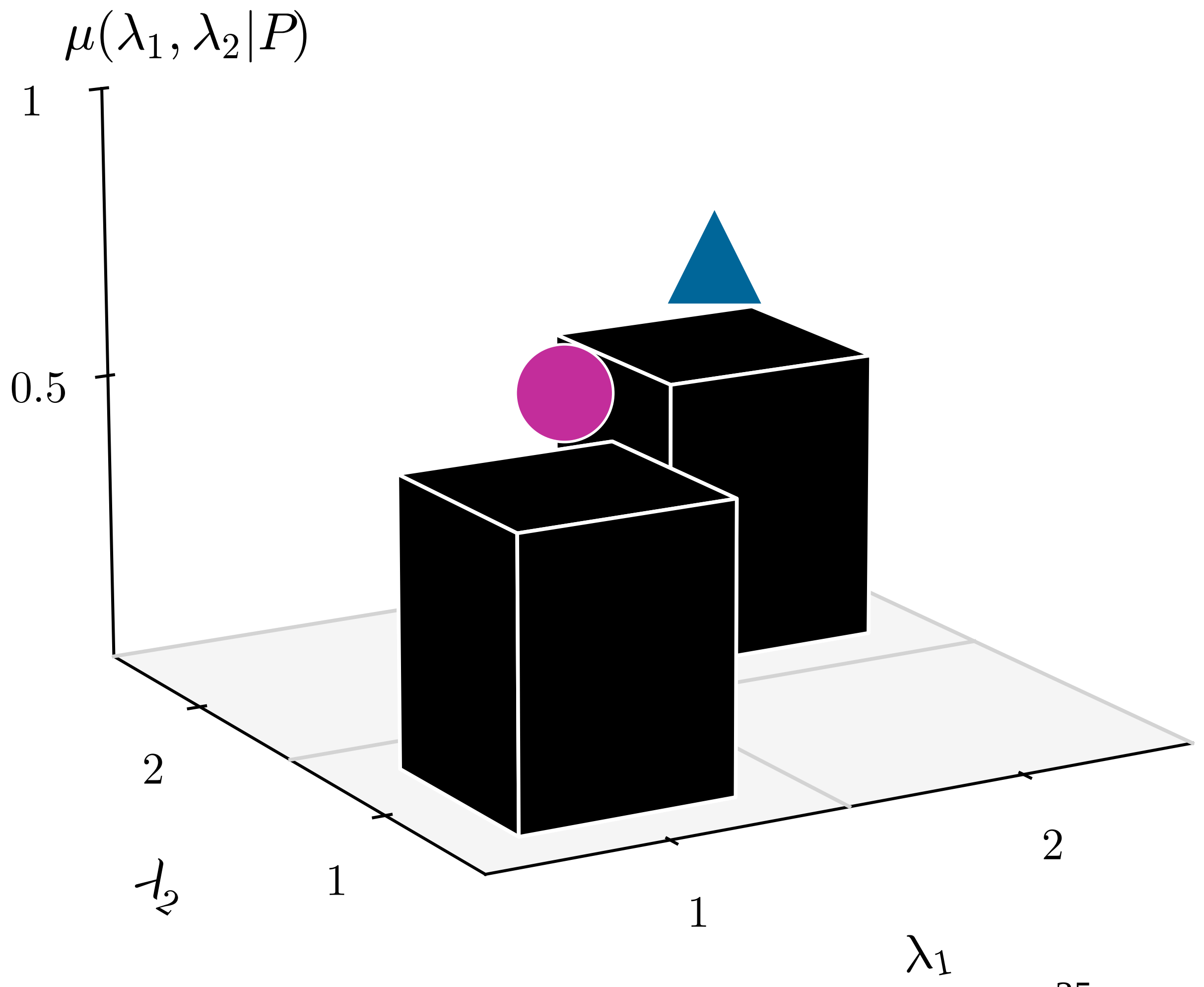




A classical experiment

Probability distribution: $\mu(\lambda|P)$





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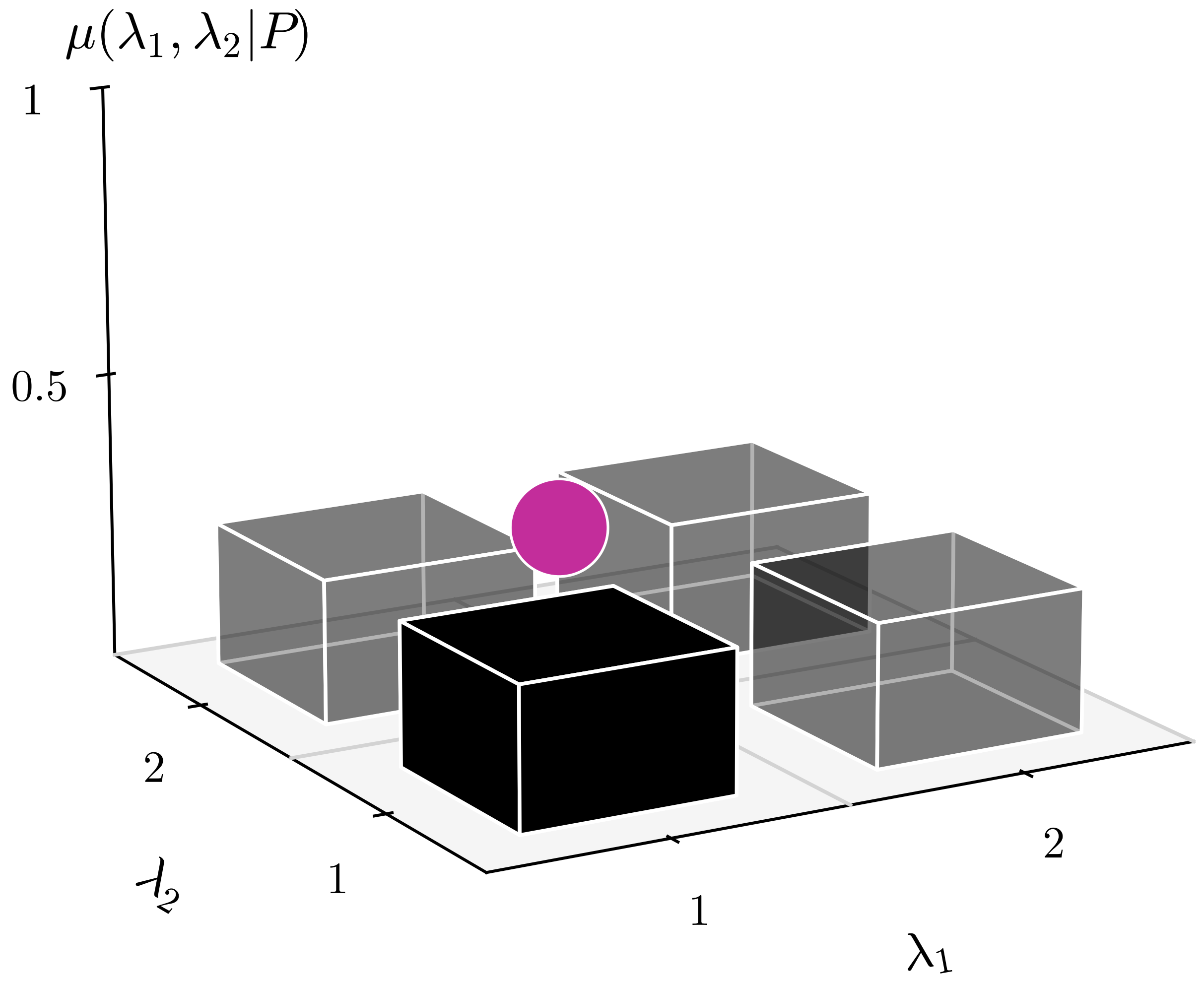
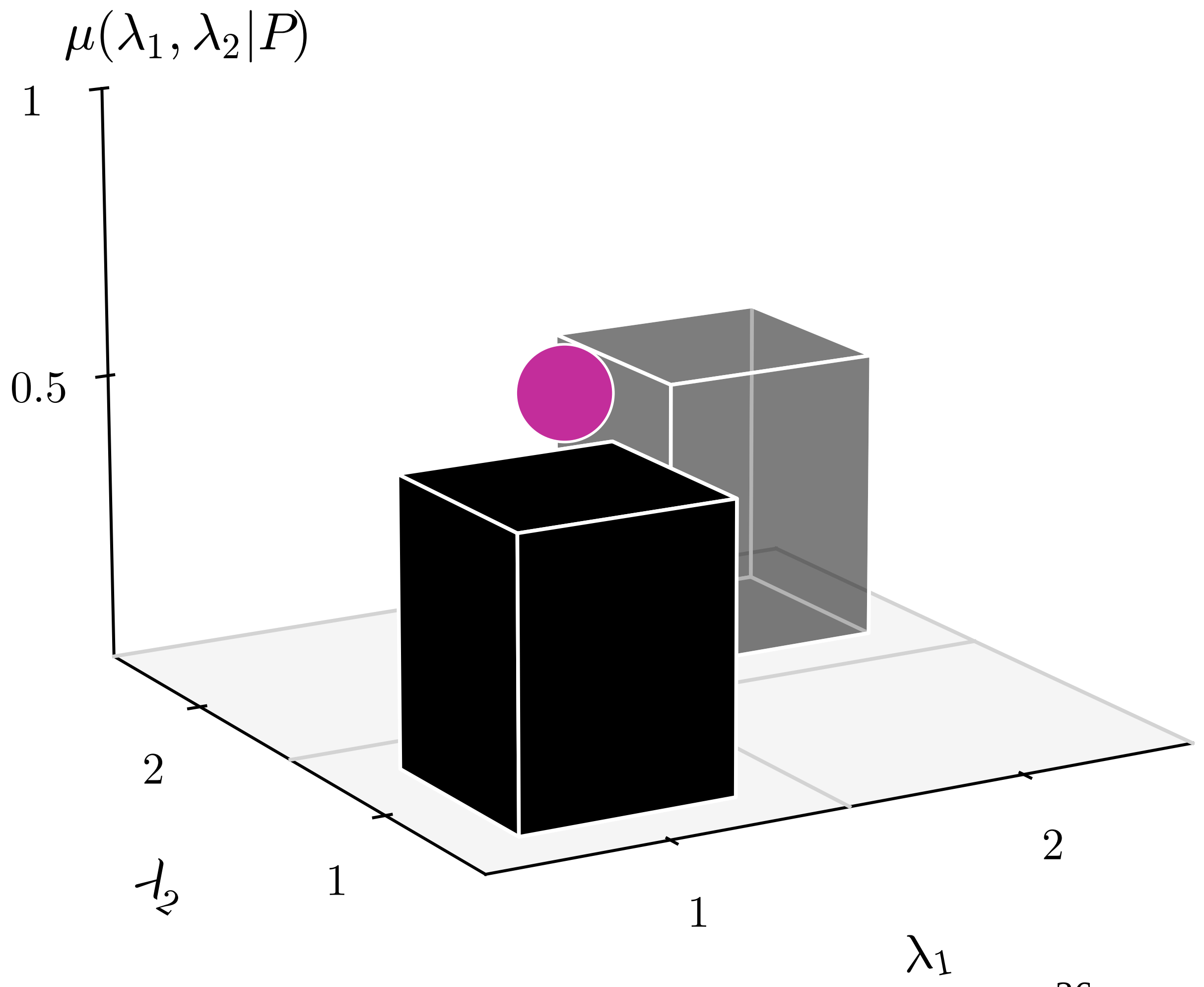
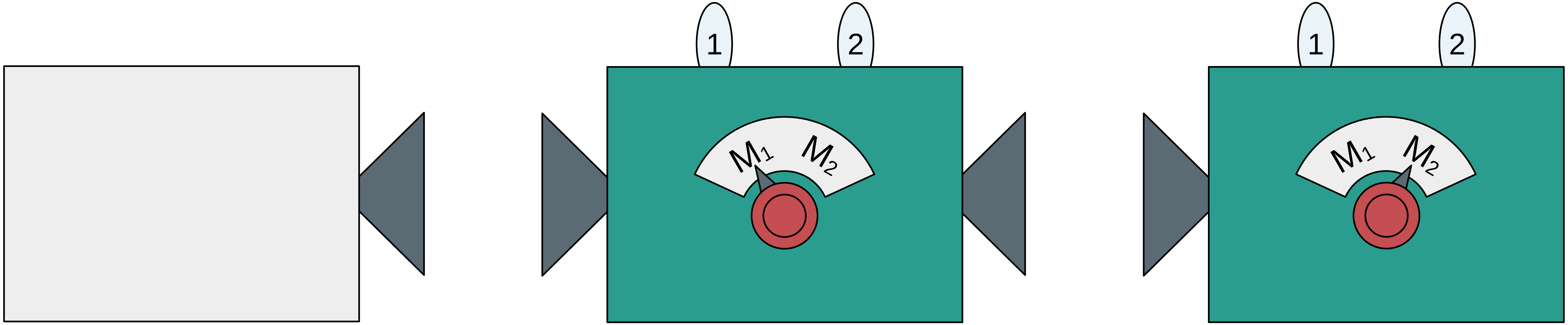




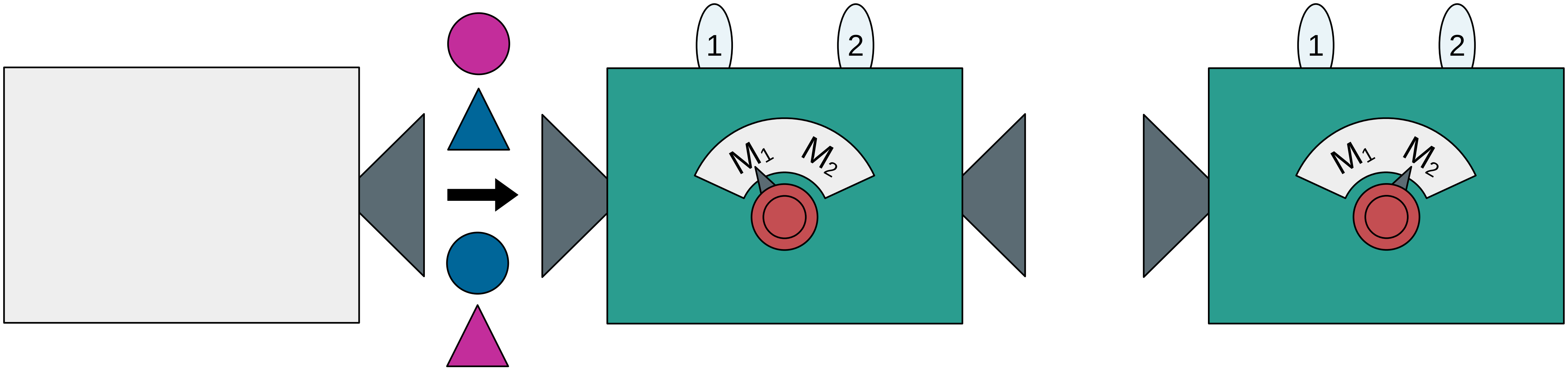
A classical experiment

Sequential (or simultaneous) measurement



A classical experiment

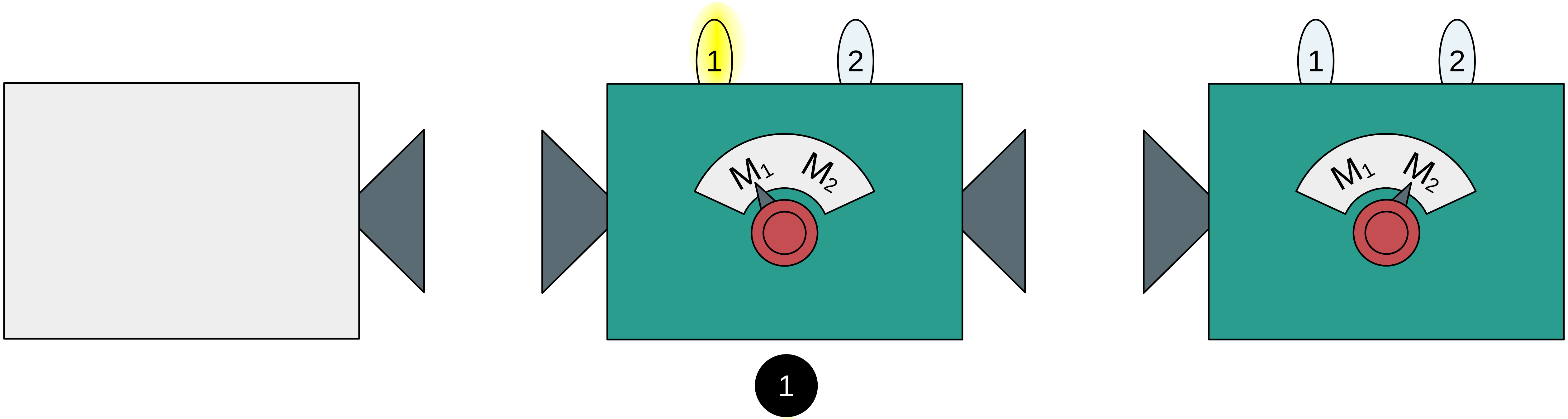
Sequential (or simultaneous) measurement



$$\lambda = (\lambda_1, \lambda_2)$$

A classical experiment

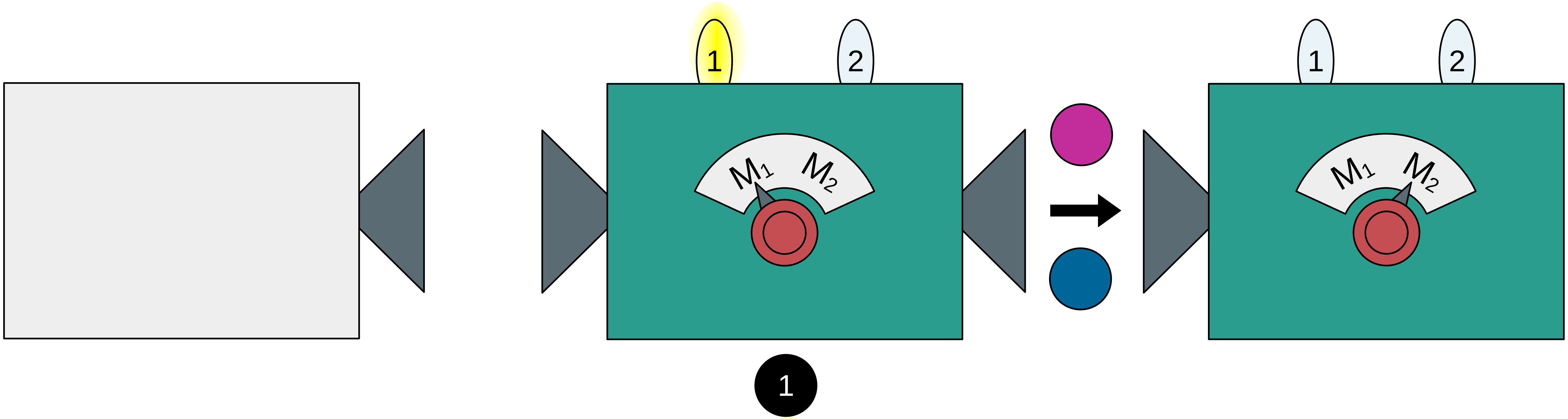
Sequential (or simultaneous) measurement



$$\lambda = (\lambda_1, \lambda_2)$$

A classical experiment

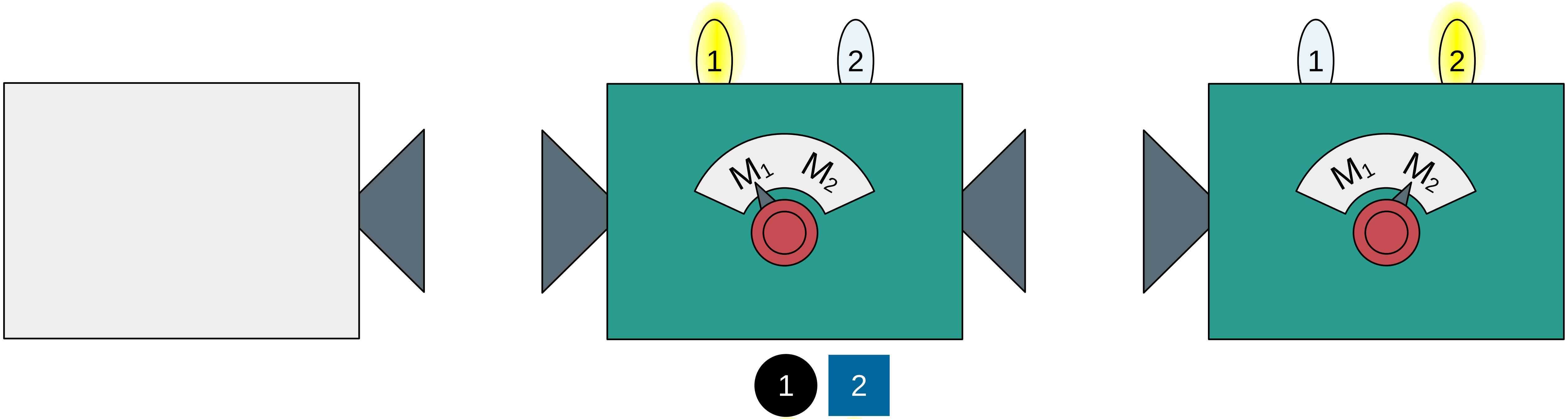
Sequential (or simultaneous) measurement



$$\lambda = (\lambda_1, \lambda_2)$$

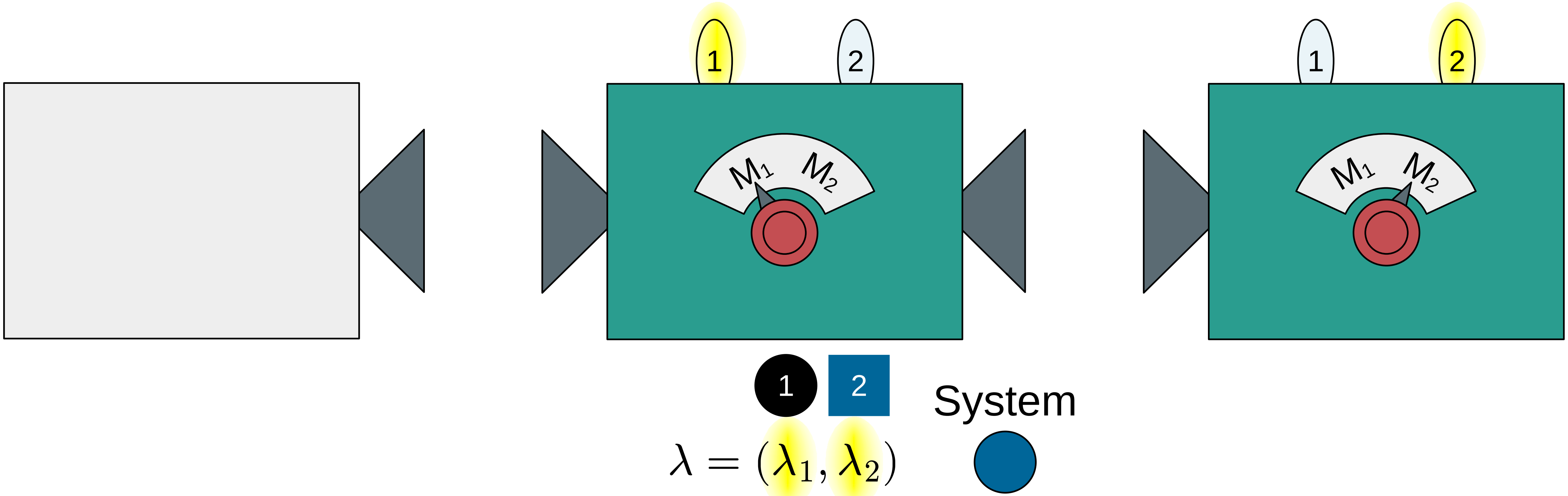
A classical experiment

Sequential (or simultaneous) measurement



$$\lambda = (\lambda_1, \lambda_2)$$

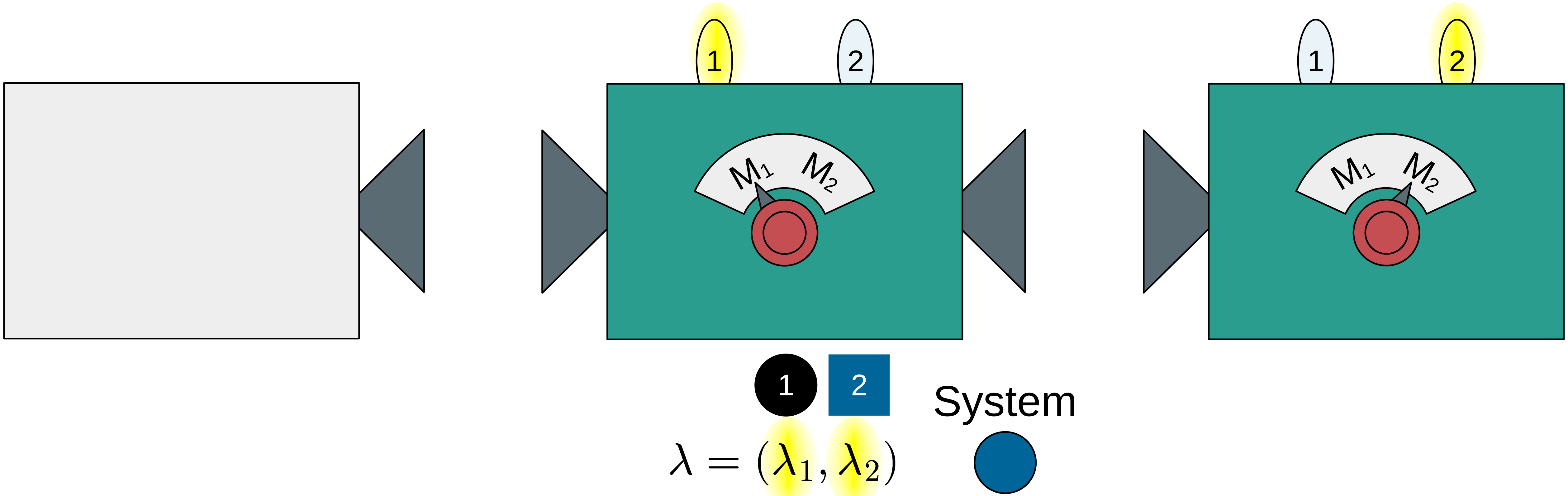
Sequential (or simultaneous) measurement



$$\lambda = (\lambda_1, \lambda_2)$$

A classical experiment

Sequential (or simultaneous) non-disturbing measurement



$\lambda = (\lambda_1, \lambda_2)$

A classical experiment

Outcome probabilities for non-disturbing measurements:

$$\mathrm{Prob}(l, m, \ldots | P, M_i, M_j, \ldots) = \sum_{\lambda} \mu(\lambda|P)\xi(l|M_i, \lambda)\xi(m|M_j, \lambda) \ldots$$

A classical experiment

Outcome probabilities for non-disturbing measurements:

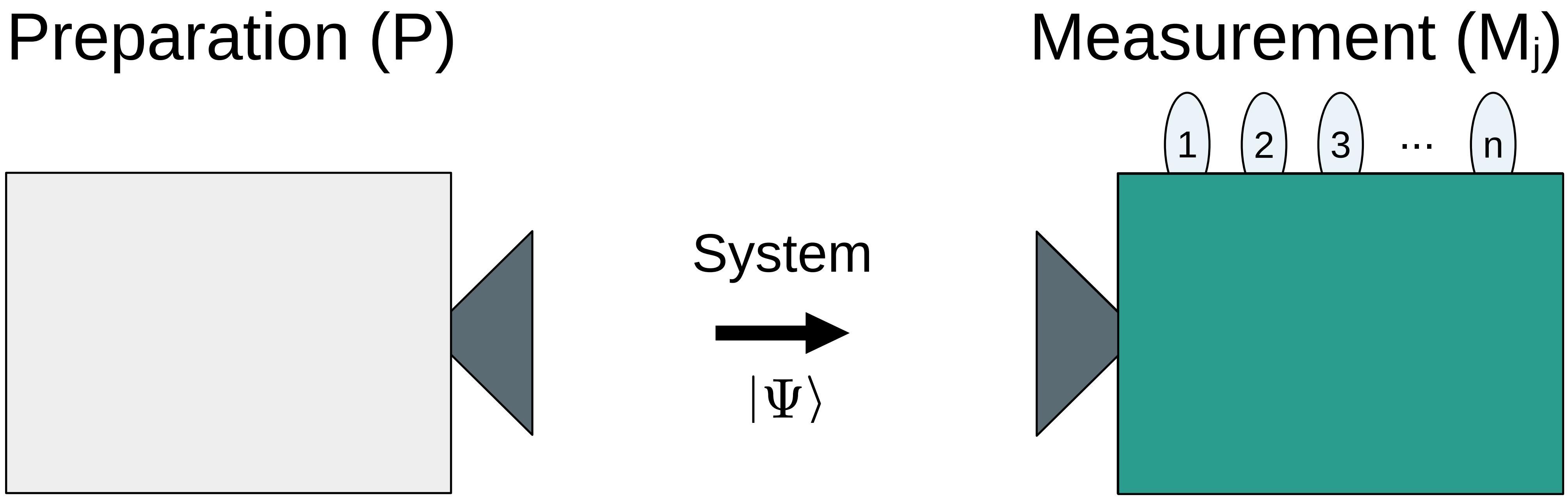
$$\mathrm{Prob}(\underbrace{l, m, ...}_{k}|P, \underbrace{M_i, M_j, ...}_{M}) = \sum_{\lambda} \mu(\lambda|P)\underbrace{\xi(l|M_i, \lambda)\xi(m|M_j, \lambda)\,...}_{\xi(k|M,\lambda)}$$

A classical experiment

Outcome probabilities for non-disturbing measurements:

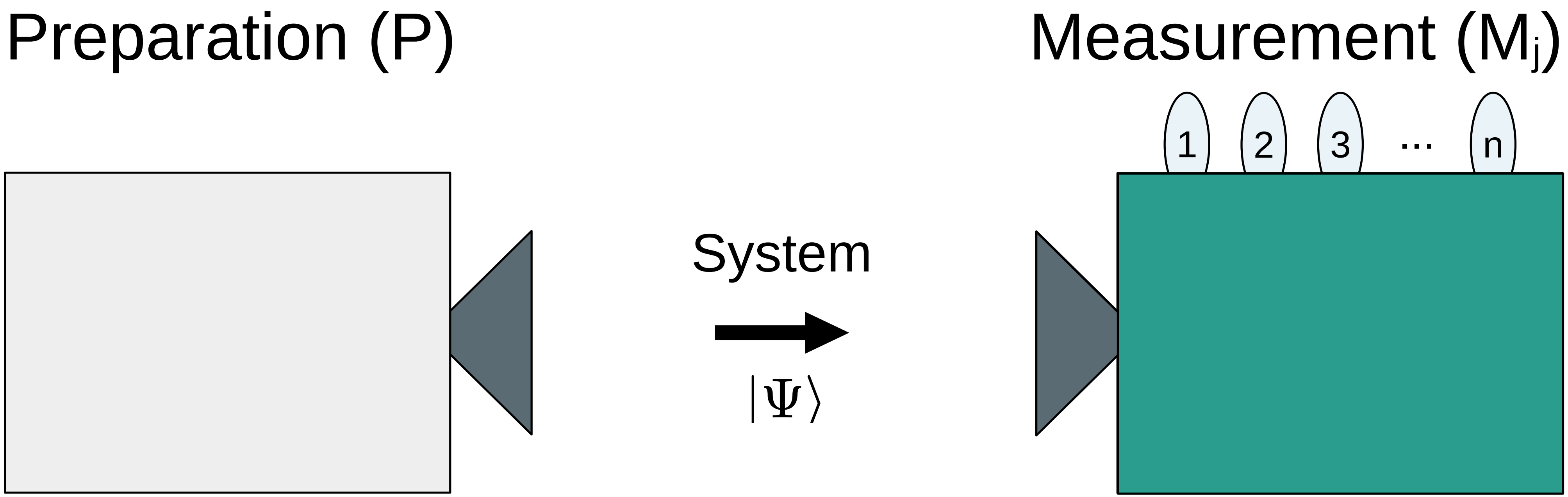
$$\mathrm{Prob}(k|P,M) = \sum_{\lambda} \mu(\lambda|P)\xi(k|M,\lambda)$$

A quantum experiment



Set of undefined properties: $|\Psi\rangle = (|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, \ldots)$

A quantum experiment

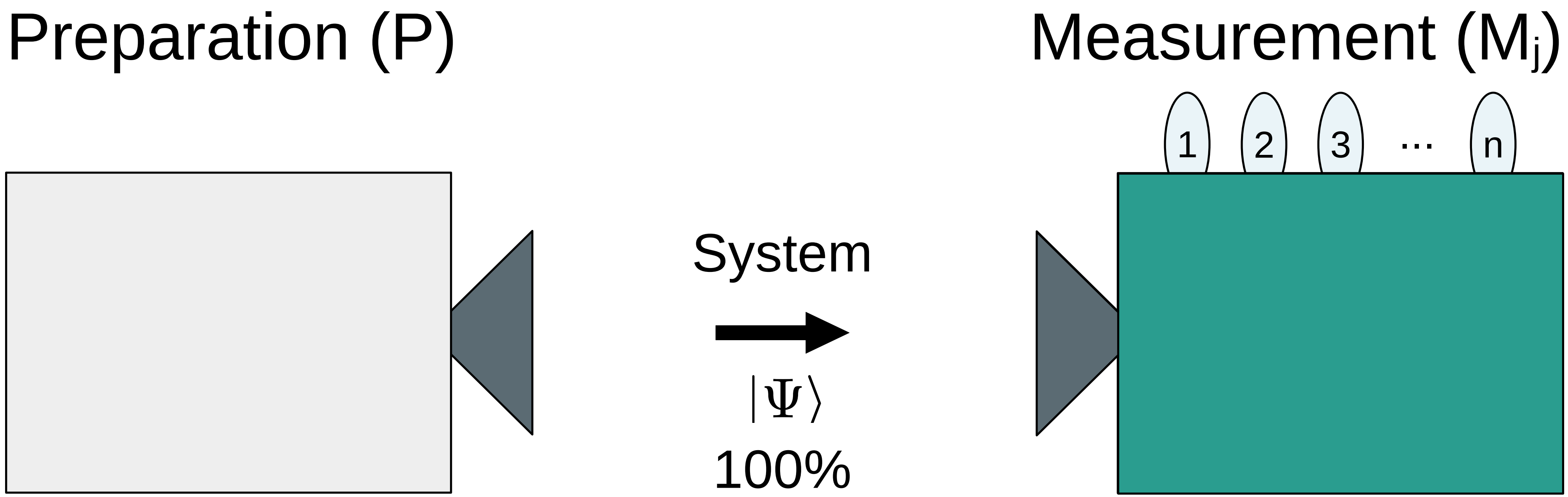


Set of undefined properties: $|\Psi\rangle = (|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, \ldots)$

Superposition: $|\psi_j\rangle = a_1|1\rangle + a_2|2\rangle + a_3|3\rangle + \ldots$

A quantum experiment

Pure state: $|\Psi\rangle$



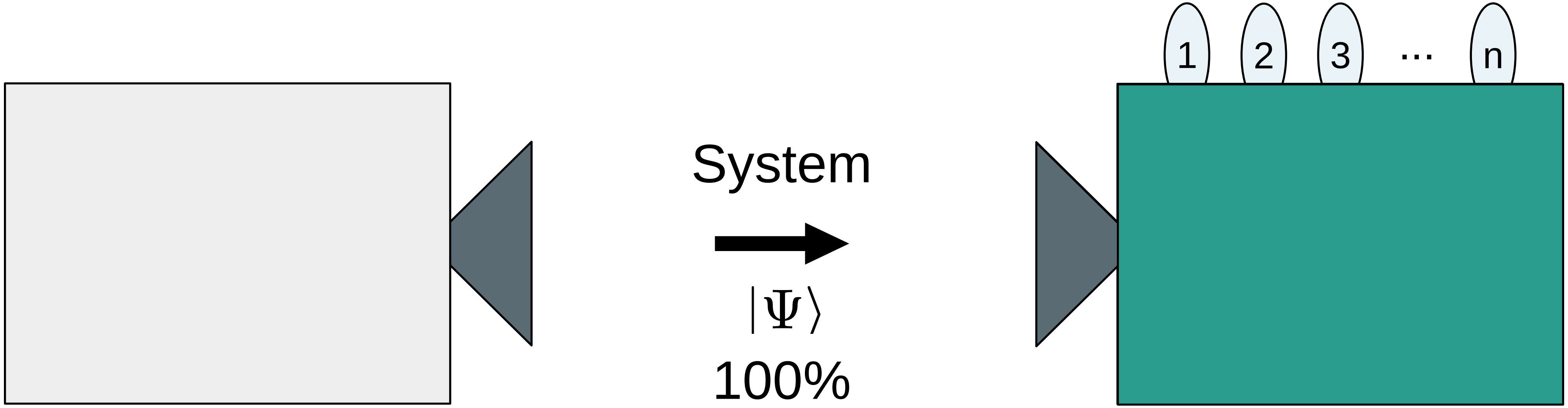
A quantum experiment

Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Preparation (P)

Measurement (M_j)



A quantum experiment

Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Outcome probabilities: $\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle \leq 1$

A quantum experiment

Pure state: $|\Psi\rangle$

Projective measurement: $\hat{\Pi}_{k|M}$

Preparation (P)

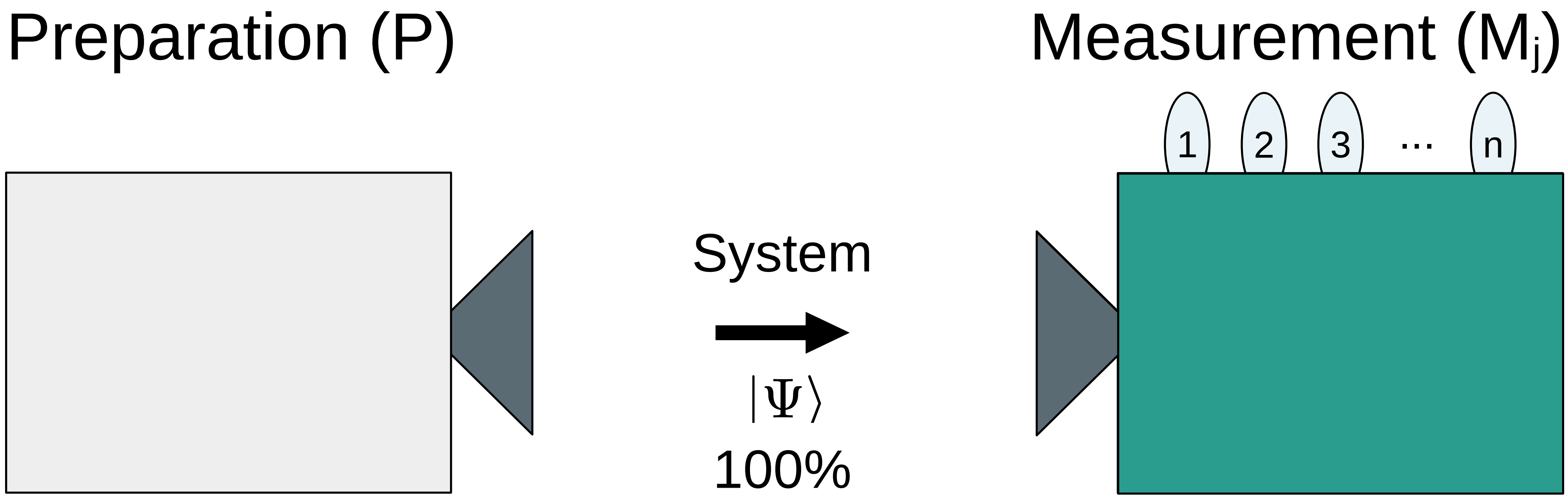
Measurement (M_j)

Is quantum mechanics secretly classical?

Outcome probabilities: $\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle \leq 1$

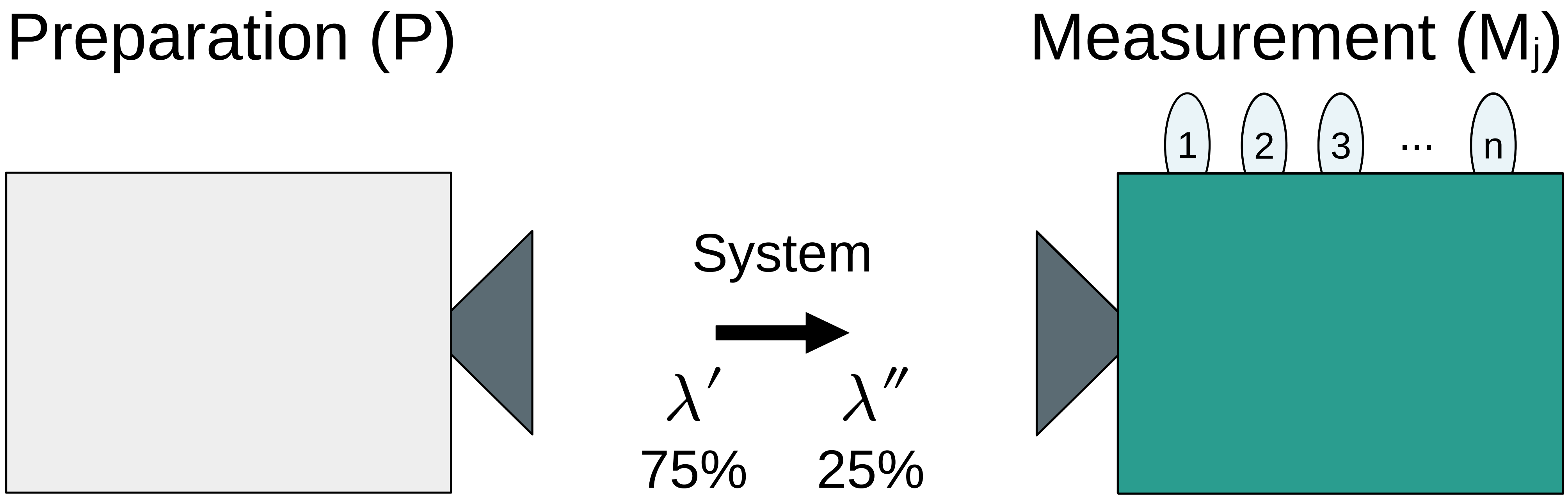
Is quantum mechanics classical?

Pure state: $|\Psi\rangle$



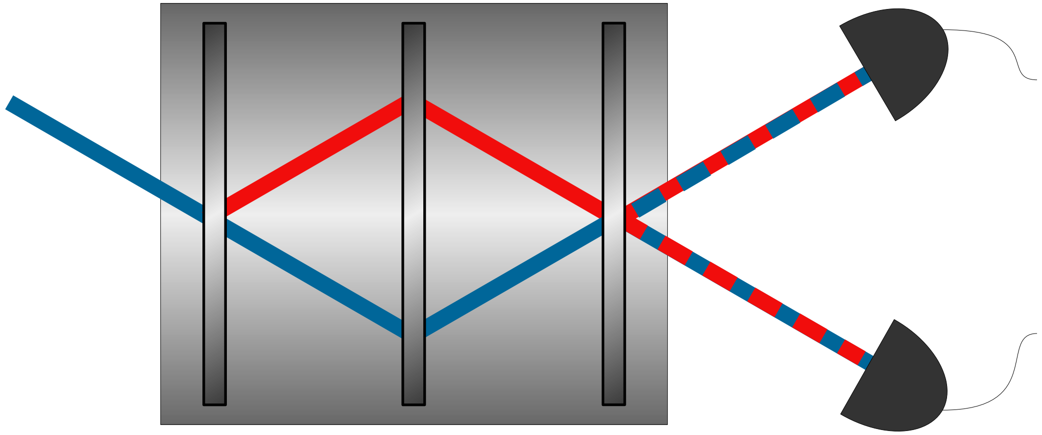
Is quantum mechanics classical?

Probability distribution: $\mu(\lambda|P_i)$



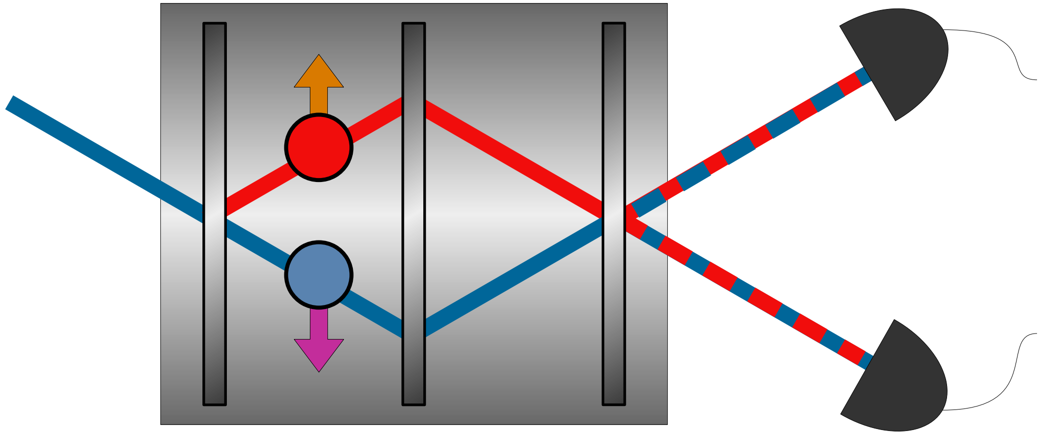
Neutron interferometry test of nonclassicality

Neutron
Interferometer:

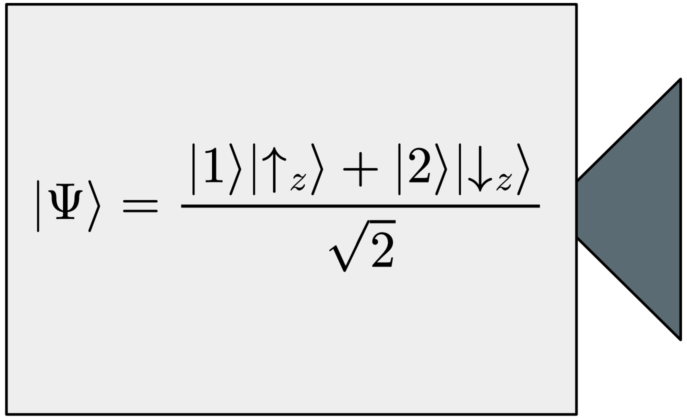


Neutron interferometry test of nonclassicality

Neutron Interferometer:



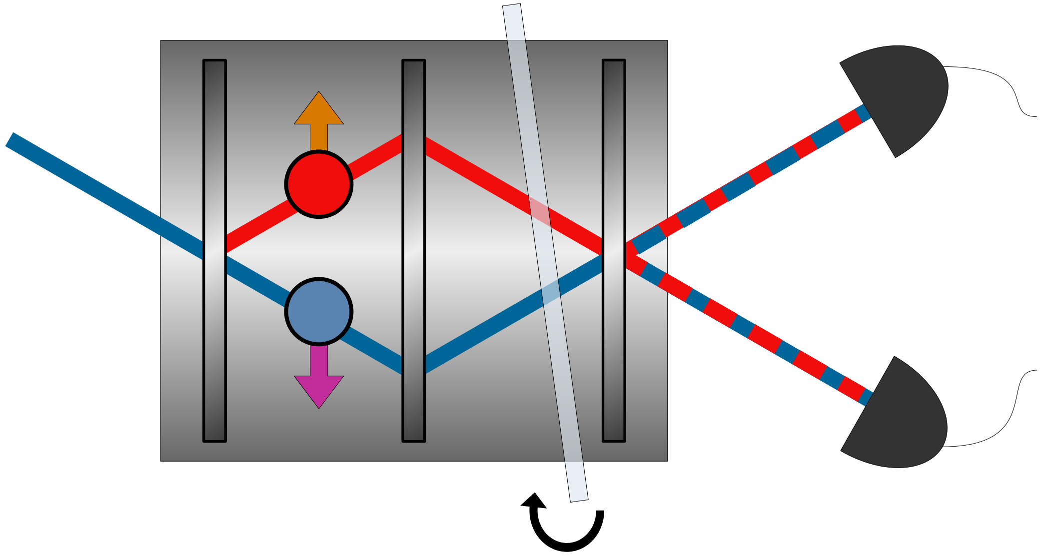
$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

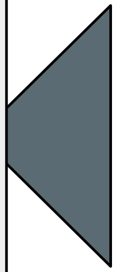


Preparation (P)

Neutron interferometry test of nonclassicality

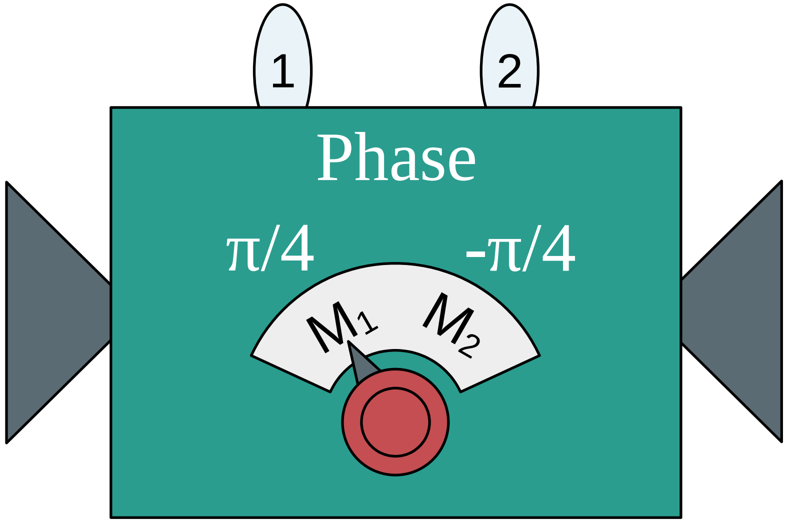
Neutron Interferometer:





$$|\Psi\rangle = \frac{|1\rangle|\uparrow_z\rangle + |2\rangle|\downarrow_z\rangle}{\sqrt{2}}$$

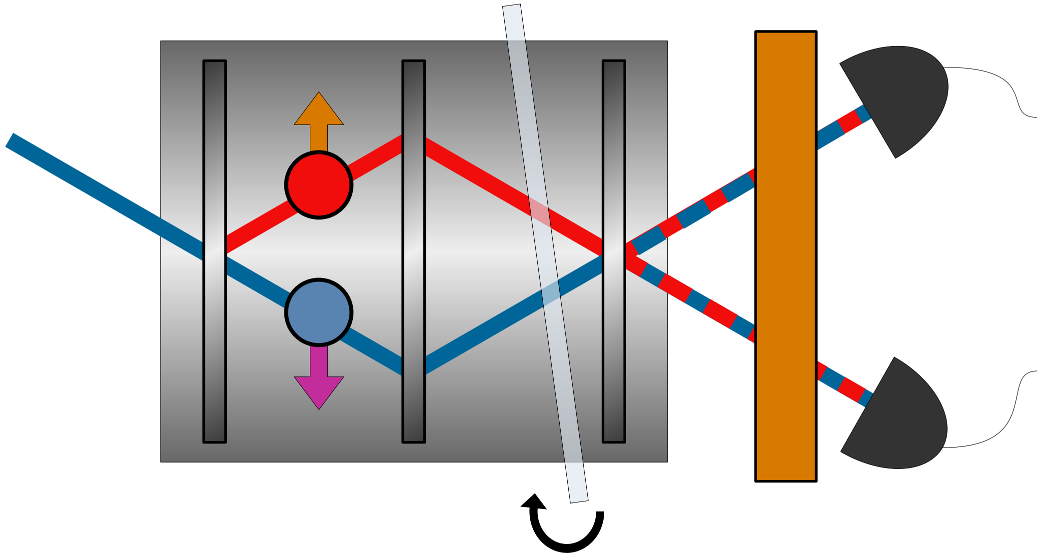
Preparation (P)

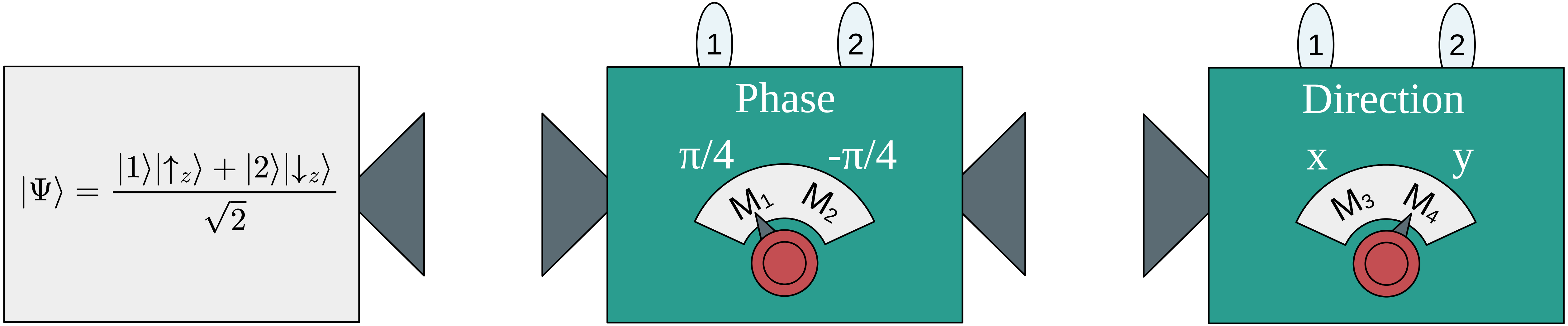


Path (M_1,M_2)

Neutron interferometry test of nonclassicality

Neutron Interferometer:



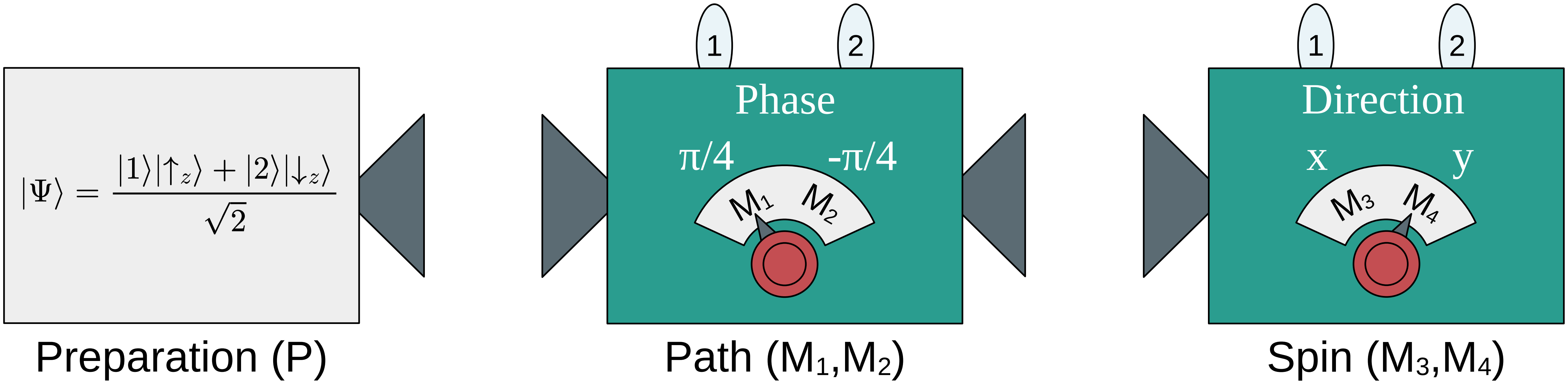


Preparation (P) Path (M_1,M_2) Spin (M_3,M_4)

Measurements:

M_1 M_2

M_3 M_4



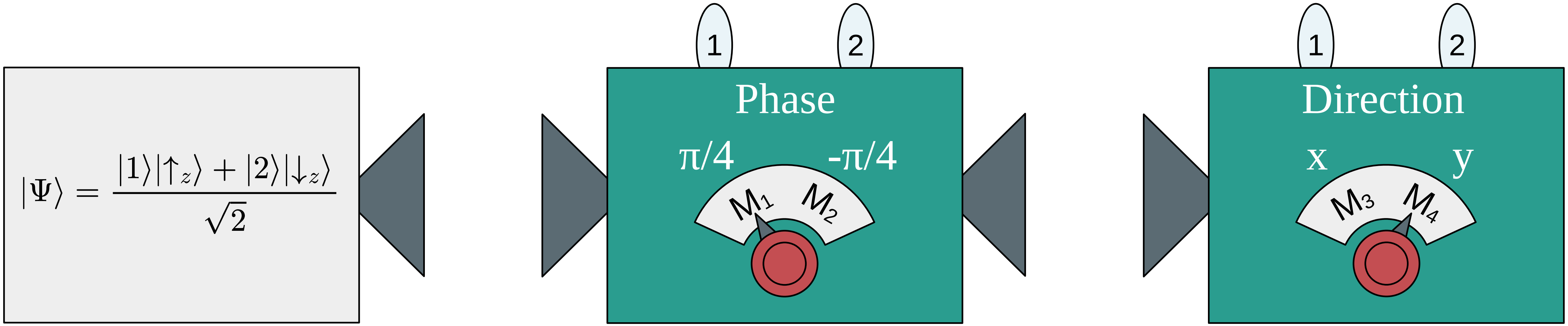
Neutron interferometry test of nonclassicality

Measurements:

$M_1 \longleftrightarrow M_2$

$M_3 \longleftrightarrow M_4$

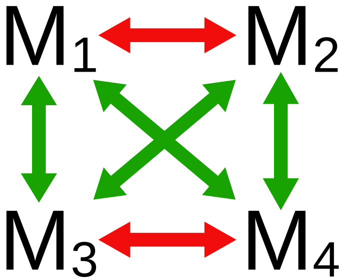
Disturbing $\longleftrightarrow$ $\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] \neq 0$



Preparation (P)   Path (M_1,M_2)   Spin (M_3,M_4)

Neutron interferometry test of nonclassicality

Measurements:



Disturbing

Non-disturbing

$$\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] \neq 0$$

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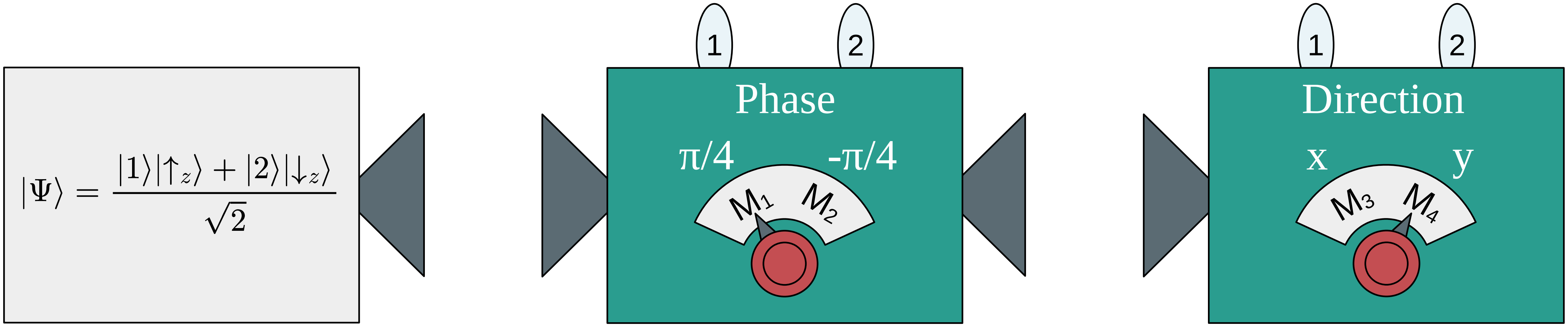

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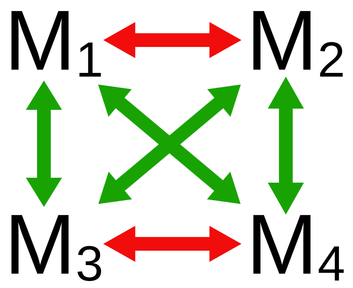

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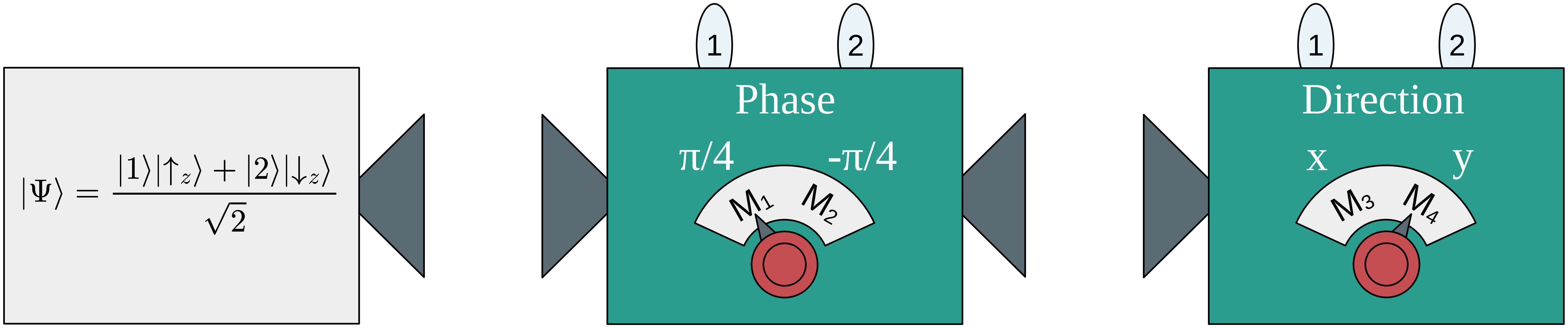

Preparation (P)

Path (M_1,M_2)

Spin (M_3,M_4)

Noncontextuality

Kochen, S., Specker, E.P. (1990). The Problem of Hidden Variables in Quantum Mechanics.

Neutron interferometry test of nonclassicality

Outcome probabilities:

$$\mathrm{Prob}(k, l|P, M_i, M_j) = \langle\Psi|\hat{\Pi}_{k|M_i}\hat{\Pi}_{l|M_j}|\Psi\rangle$$

$$\stackrel{?}{=} \sum_{\lambda} \mu(\lambda|P)\xi(k|M_i,\lambda)\xi(l|M_j,\lambda)$$

Neutron interferometry test of nonclassicality

Outcomes correlations:

$$E(M_i, M_j) = \mathrm{Prob}(k, k|P, M_i, M_j) - \mathrm{Prob}(k, l \neq k|P, M_i, M_j)$$

Clauser–Horne–Shimony–Holt (CHSH) inequality:

$$E(M_1, M_3) - E(M_1, M_4) + E(M_2, M_3) + E(M_2, M_4) \leq 2$$

Neutron interferometry test of nonclassicality

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Eperimental results:

$$2.051 \pm 0.019$$

Hasegawa, Y., Loidl, R., Badurek, G., Baron, M. & Rauch, H. Violation of a bell-like inequality in single-neutron interferometry. Nature 425, 45–48 (2003).

Neutron interferometry test of nonclassicality

Is quantum mechanics secretly classical?

...NO!

Neutron interferometry test of nonclassicality

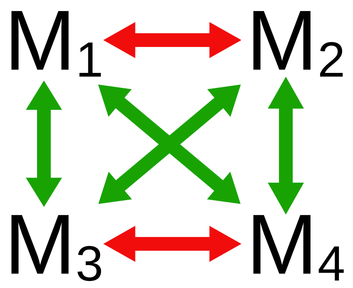
Is quantum mechanics secretly classical?

...NO!*

*according to our definition of non-disturbance (commutativity)

Bell test of nonclassicality

Measurements:



Disturbing

Non-disturbing

$$\left[\hat{\Pi}_{k|M_i}, \hat{\Pi}_{l|M_j}\right] \neq 0$$

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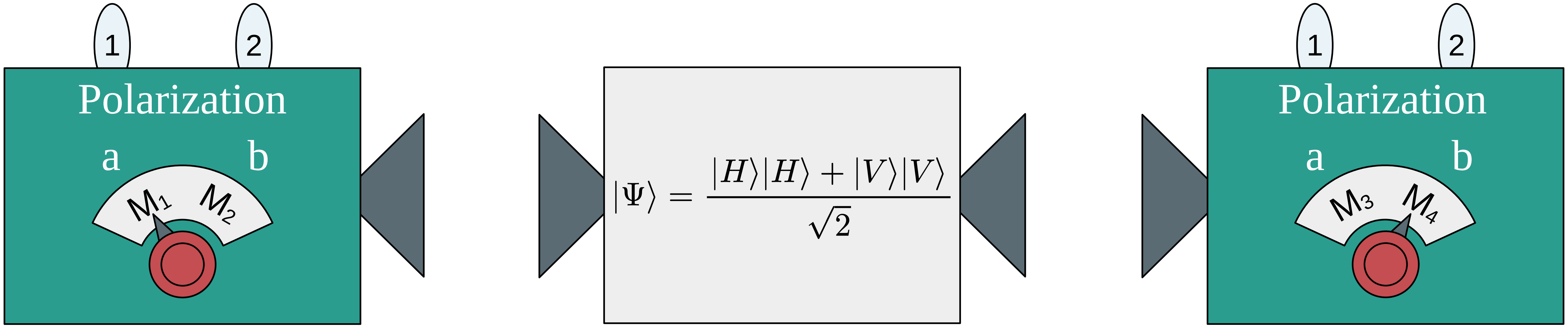

Particle 1 (M_1,M_2)

Preparation (P)

Particle 2 (M_3,M_4)

Bell test of nonclassicality

Is quantum mechanics secretly classical?

Still...NO!*

*independently of the quantum formulation

Quasiprobability representations and weak values

Is quantum mechanics classical?

Outcome probabilities:

$$\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle \overset{?}{=} \sum_{\lambda} \mu(\lambda|P)\xi(k|M,\lambda)$$

Quasiprobability representations

Two orthonormal basis: $\{|a_i\rangle\}, \{|b_j\rangle\}$

Kirkwood-Dirac quasiprobability representation:

$$\mathrm{Prob}(k|P,M) = \langle\Psi|\hat{\Pi}_{k|M}|\Psi\rangle = \sum_{a_i,b_j} \langle\Psi|b_j\rangle\langle b_j|a_i\rangle\langle a_i|\Psi\rangle \, \frac{\langle b_j|\hat{\Pi}_{k|M}|a_i\rangle}{\langle b_j|a_i\rangle}$$

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$$= \sum_{\lambda=a_i,b_j} \tilde{\mu}(\lambda|P)\tilde{\xi}(k|\lambda,M)$$

Quasiprobability representations

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$$= \sum_{\lambda=a_i,b_j} \tilde{\mu}(\lambda|P)\tilde{\xi}(k|\lambda,M)$$

Anomalous features:

$$\tilde{\mu}(\lambda|\Psi), \tilde{\xi}(k|\lambda,\hat{\Pi}_{k|M}) < 0$$

$$\tilde{\xi}(k|\lambda,\hat{\Pi}_{k|M}) > 1$$

$$\tilde{\mu}(\lambda|\Psi), \tilde{\xi}(k|\lambda,\hat{\Pi}_{k|M}) \in \mathbb{C}$$

Quasiprobability representations

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Anomalous features:

$$\tilde{\mu}(\lambda|\Psi), \tilde{\xi}(k|\lambda, \hat{\Pi}_{k|M}) < 0$$
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Weak value

..and we can measure it!

Quasiprobability representations

Two orthonormal basis: $\{|a_i\rangle\}, \{|b_j\rangle\}$

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$$= \sum_{a_i,b_j} |\langle b_j|\Psi\rangle|^2 \, \frac{\langle b_j|a_i\rangle\langle a_i|\Psi\rangle}{\langle b_j|\Psi\rangle} \, \frac{\langle b_j|\hat{\Pi}_{k|M}|a_i\rangle}{\langle b_j|a_i\rangle}$$

Outcome probabilty

Weak values!

..and we can measure them!

A standard (Von Neumann) measurement

Stern-Gerlach

Spin Position

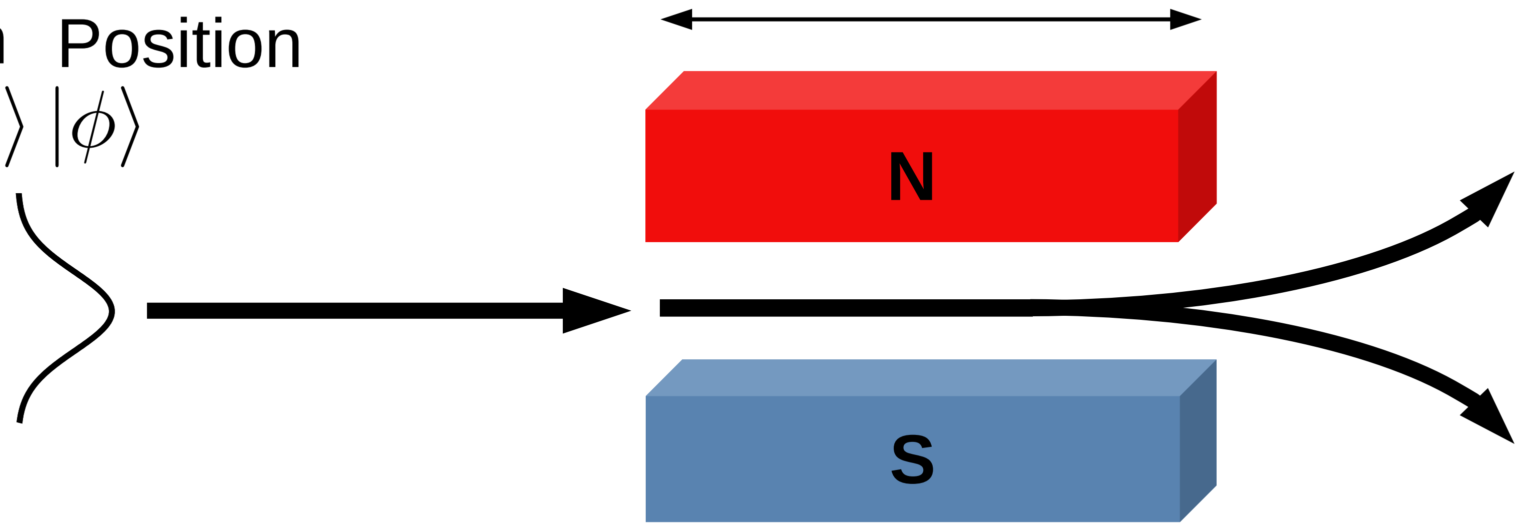
$|S_{in}\rangle \, |\phi\rangle$

A standard (Von Neumann) measurement

Stern-Gerlach

Spin Position

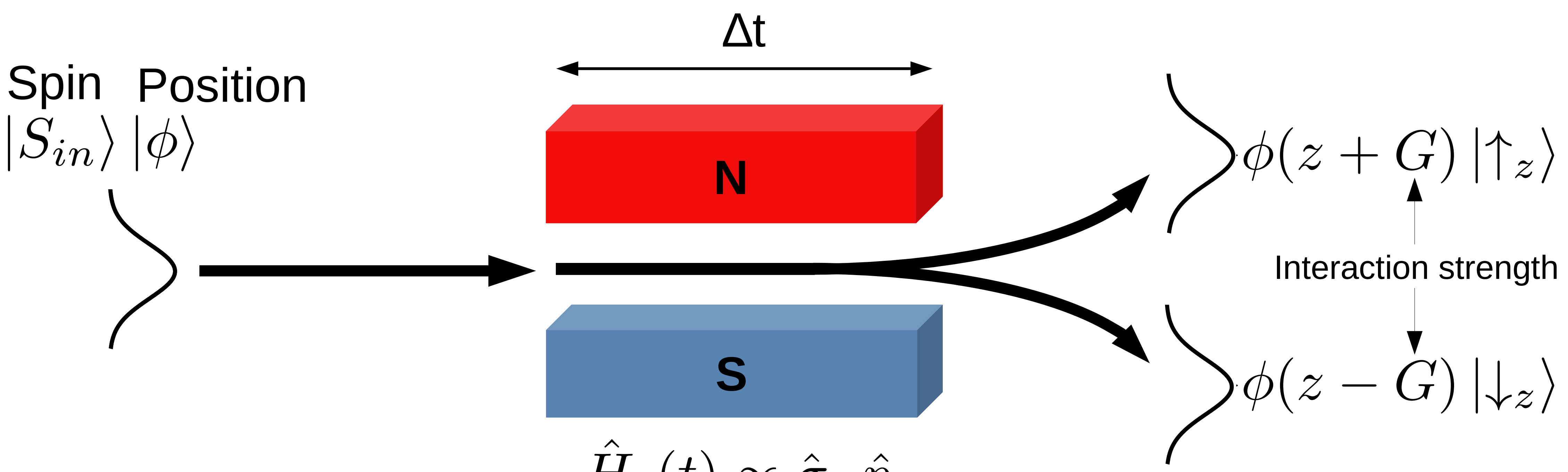
$|S_{in}\rangle\,|\phi\rangle$



$$\hat{H}_I(t) \propto \hat{\sigma}_z\,\hat{p}_z$$

A standard (Von Neumann) measurement

Stern-Gerlach

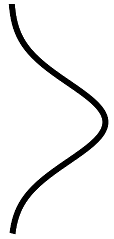


$$\hat{H}_I(t) \propto \hat{\sigma}_z\,\hat{p}_z$$

Weak values and weak measurements

Spin Position

$$|S_{in}\rangle\,|\phi\rangle$$



Aharonov, Yakir, et al. "How the result of a measurement of a component of the spin of a spin-1/2 particle can turn out to be 100." Physical review letters 60.14 (1988): 1351.

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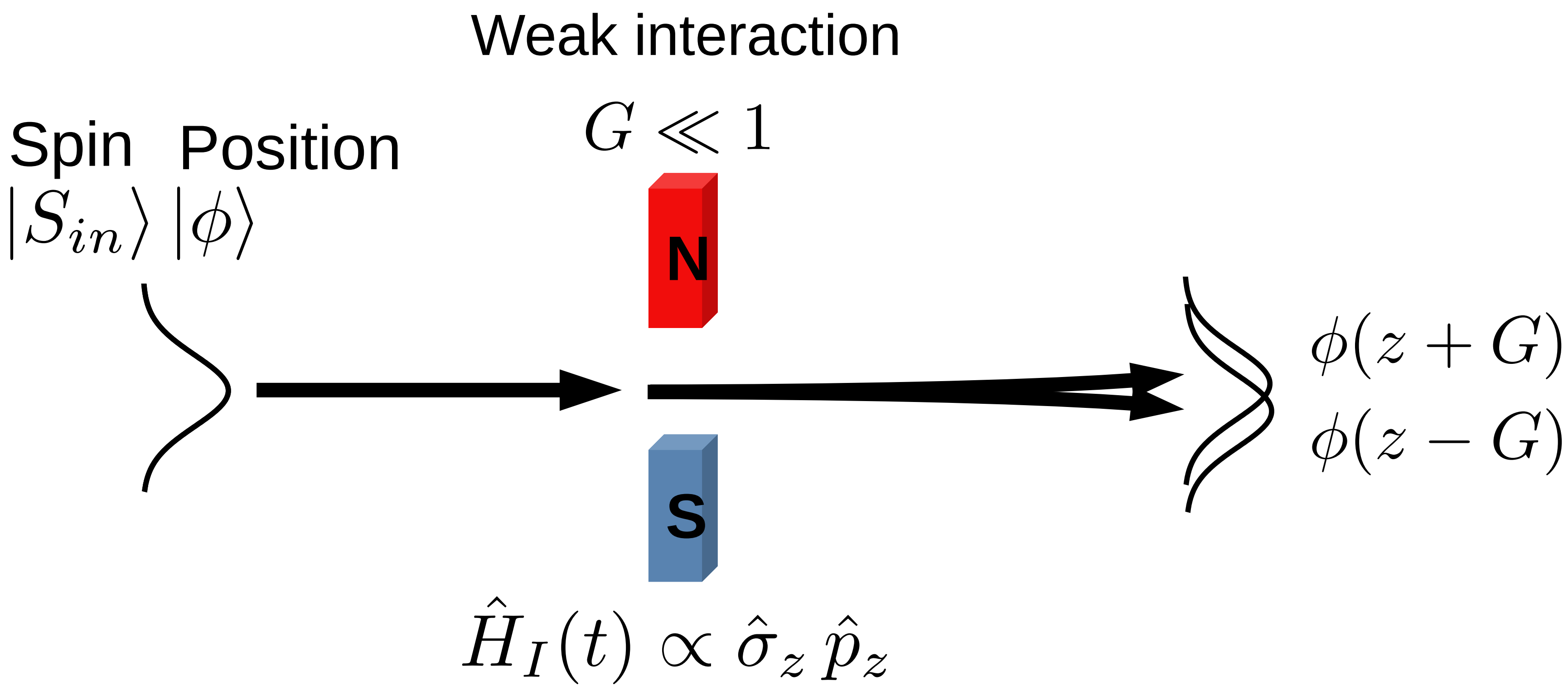

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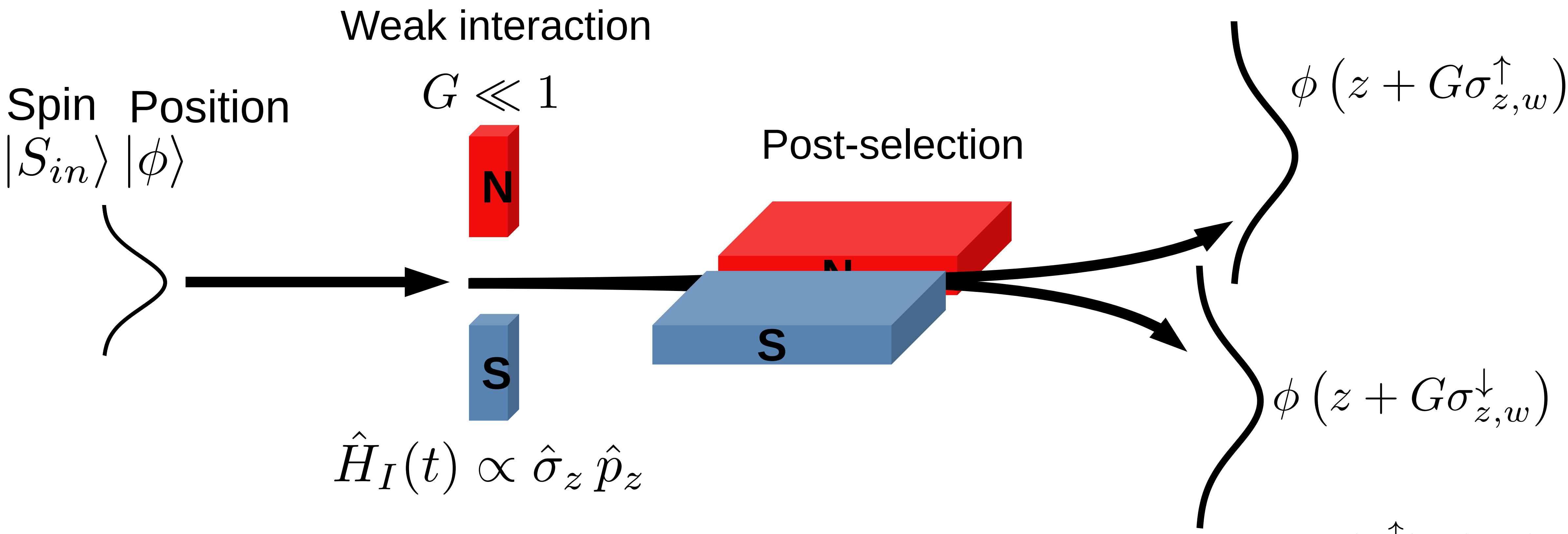

Spin weak value: $\sigma^{\updownarrow}_{z,w} = \dfrac{\langle S^{\updownarrow}_{\mathrm{f}}|\hat{\sigma}_z|S_{\mathrm{in}}\rangle}{\langle S^{\updownarrow}_{\mathrm{f}}|S_{\mathrm{in}}\rangle}$

Aharonov, Yakir, et al. "How the result of a measurement of a component of the spin of a spin-1/2 particle can turn out to be 100." Physical review letters 60.14 (1988): 1351.

Experimental results

Weak values extraction in neutron interferometry

Meter state and weak interaction:

- *Lemmel, Hartmut, et al. "Quantifying the presence of a neutron in the paths of an interferometer." Physical Review Research 4.2 (2022): 023075.*
- *Danner, Armin, et al. "Simultaneous path weak-measurements in neutron interferometry." Scientific Reports 14.1 (2024): 25994.*
- *Dvorak, Andreas, et al. "Tight qubit uncertainty relations studied through weak values in neutron interferometry." Physical Review Research 7.4 (2025): 043334.*

Meter state and no weak interaction:

- *Denkmayr, Tobias, et al. "Weak values from strong interactions in neutron interferometry." Physica B: Condensed Matter 551 (2018): 339-346.*

No meter state and weak interaction:

- *Masiello, Ismaele V., et al. "Simultaneous determination of two path weak-values with time-dependent phase manipulation in neutron interferometry." New Journal of Physics 27.2 (2025): 023017.*

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Meter states or weak interactions increase the experimental complexity!

The manipulation of additional degrees of freedom increases the amount of required experimental resources (additional instrumentation and procedures), while weak interactions typically necessitate longer measurement times due to the limited information gained per detection event.

Weak values extraction in neutron interferometry

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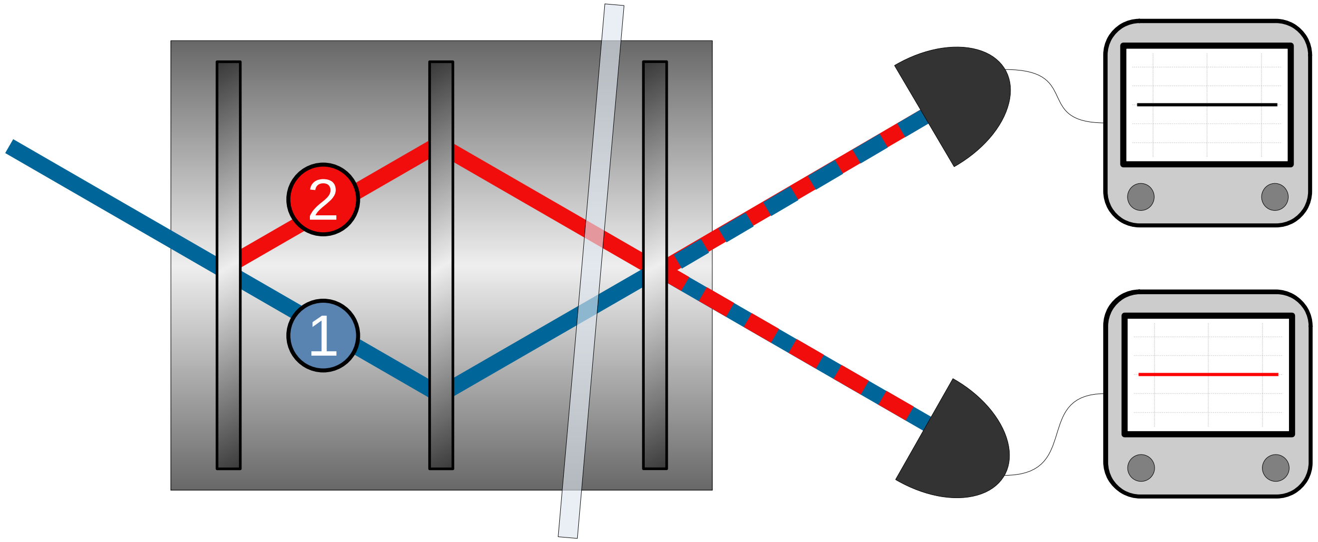
No meter state and no weak interaction:

- *Masiello, Ismaele V., et al. "Anomalous weak values in a generalized Mach-Zehnder interferometer extracted directly from intensity measurements." in review at npj Quantum Information.*

Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



Initial state:

$$|\psi_{\mathrm{in}}\rangle = \cos\left(\frac{\theta}{2}\right)|1\rangle + \mathrm{e}^{\mathrm{i}\phi}\sin\left(\frac{\theta}{2}\right)|2\rangle$$

Path observable:

$$\hat{\Pi}_1 = |1\rangle\langle 1| \qquad \hat{\Pi}_2 = |2\rangle\langle 2|$$

Weak values from output intensities

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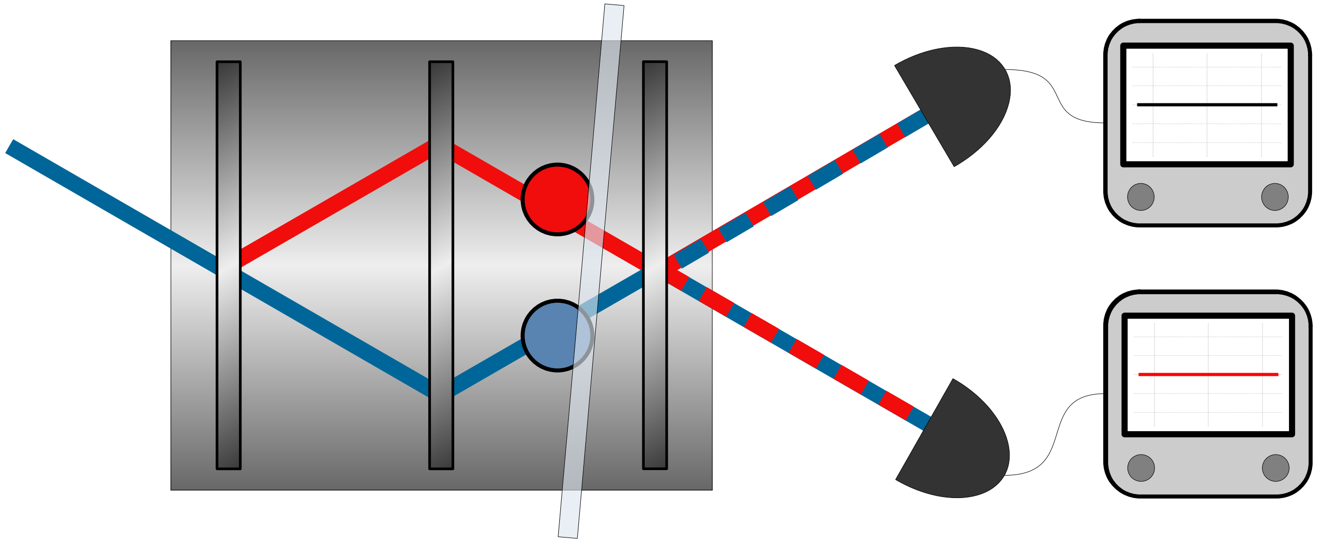

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Phase shift:

$$|\psi_{\text{in}}\rangle \to \mathrm{e}^{-\mathrm{i}\delta\hat{\Pi}_1}|\psi_{\text{in}}\rangle$$

Weak values from output intensities

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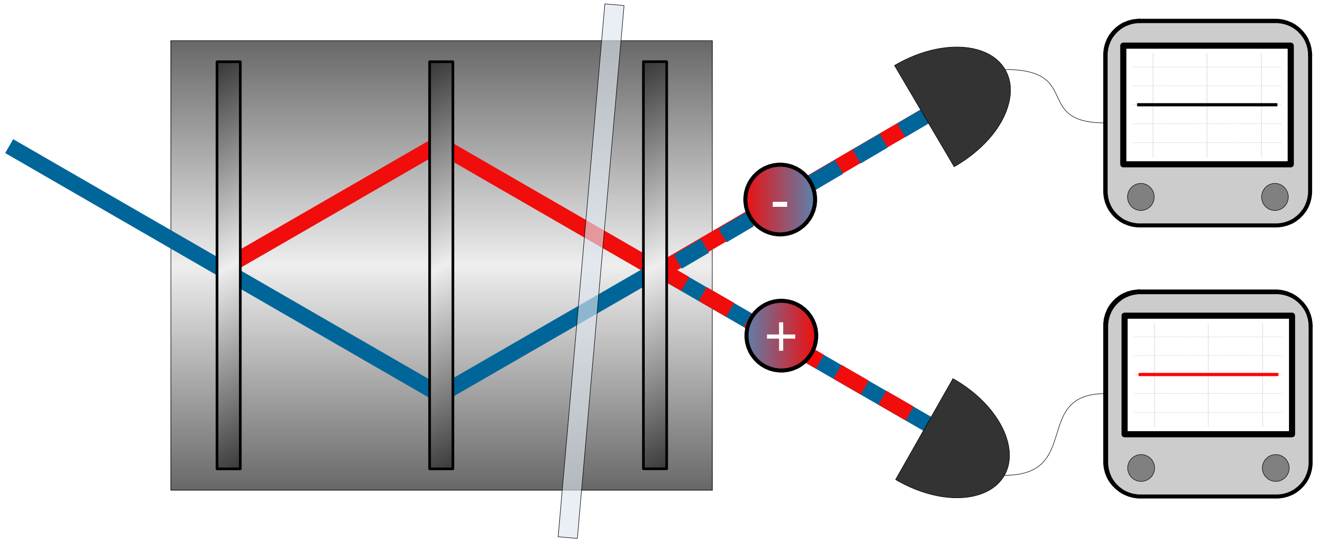

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Final states:

$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \quad |\psi_-\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}}$$

Weak values from output intensities

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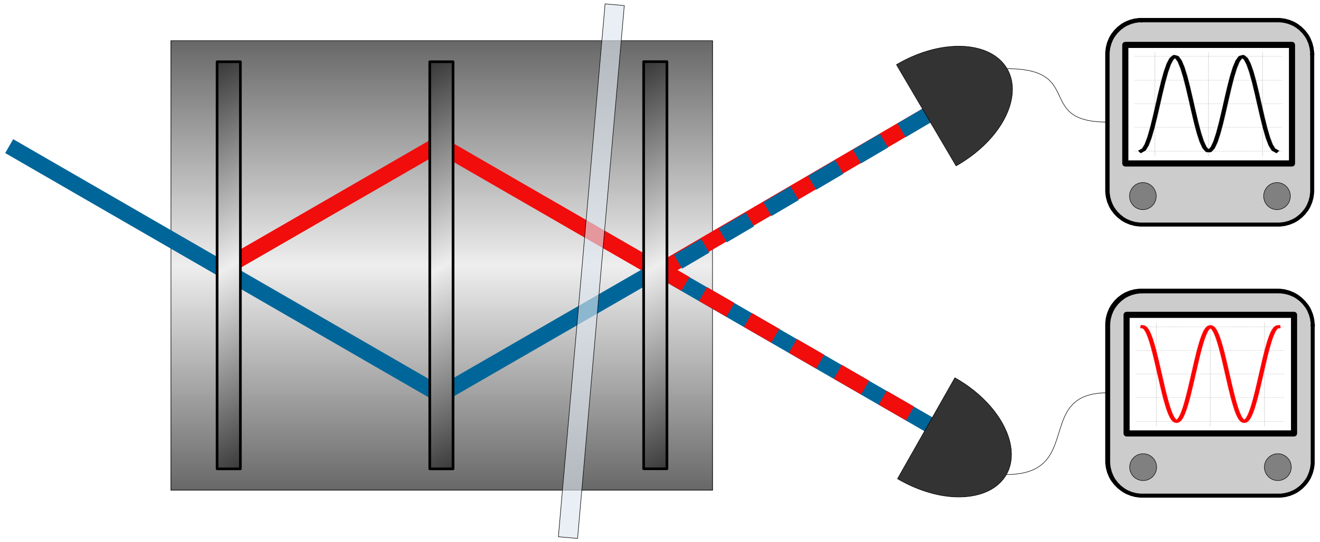

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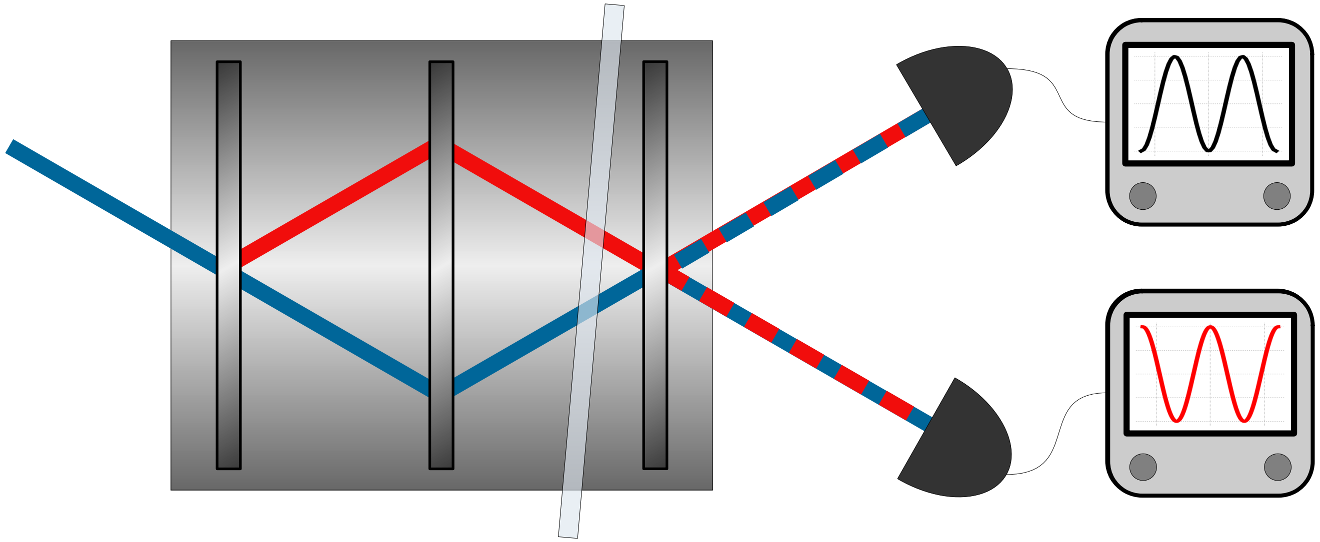
Measured intensity:

$$I_\pm(\delta) = A\left|\langle\psi_\pm|\mathrm{e}^{-\mathrm{i}\delta\hat{\Pi}_1}|\psi_{\text{in}}\rangle\right|^2$$

Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



Path weak-values:

$$w_{\pm,j} = \frac{\langle\psi_\pm|\hat{\Pi}_j|\psi_{\text{in}}\rangle}{\langle\psi_\pm|\psi_{\text{in}}\rangle} = w^{\text{R}}_{\pm,j} + \text{i}\, w^{\text{I}}_{\pm,j}$$

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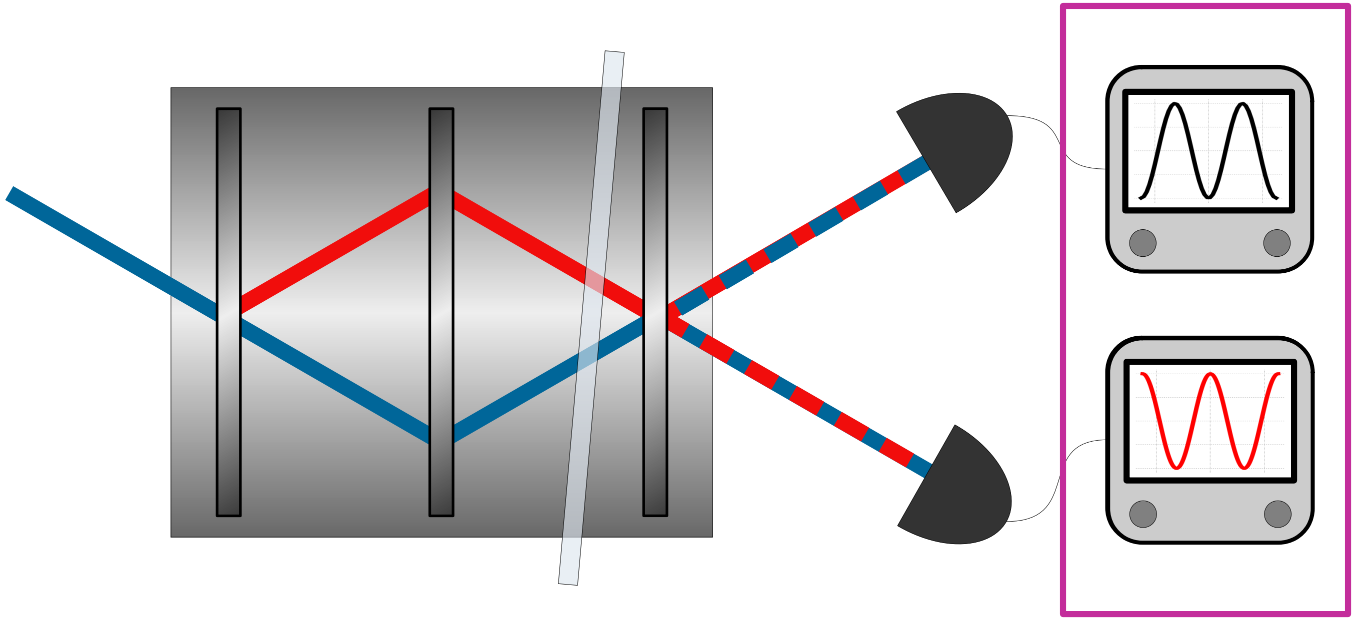
Weak values from output intensities

Generalized neutron Mach-Zehnder interferometer:



Weak values can be directly extracted from the output intensities!

Path weak-values:

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Weak values from output intensities

Measured intensity:

$$
\begin{aligned}
I_{\pm}(\delta) &= A \left| \langle \psi_{\pm} | \mathrm{e}^{-\mathrm{i}\delta \hat{\Pi}_1} | \psi_{\mathrm{in}} \rangle \right|^2 \\
&= A \left| \langle \psi_{\pm} | \left[1 + \left(\mathrm{e}^{-\mathrm{i}\delta} - 1 \right) \hat{\Pi}_1 \right] | \psi_{\mathrm{in}} \rangle \right|^2 \\
&= A \left| \langle \psi_{\pm} | \psi_{\mathrm{in}} \rangle \left[1 + \left(\mathrm{e}^{-\mathrm{i}\delta} - 1 \right) w_{\pm,1} \right] \right|^2 \\
&= A \left| \langle \psi_{\pm} | \psi_{\mathrm{in}} \rangle \right|^2 \left[1 + 2 \left(|w_{\pm,1}|^2 - w^{\mathrm{R}}_{\pm,1} \right) (1 - \cos\delta) + 2\, w^{\mathrm{I}}_{\pm,1} \, \sin\delta \right] \\
&= A \left| \langle \psi_{\pm} | \psi_{\mathrm{in}} \rangle \right|^2 \left[1 + 2 \left(|w_{\pm,2}|^2 - w^{\mathrm{R}}_{\pm,2} \right) (1 - \cos\delta) - 2\, w^{\mathrm{I}}_{\pm,2} \, \sin\delta \right] \; ,
\end{aligned}
$$

Weak values from output intensities

Measured intensity:

$$
\begin{aligned}
I_\pm(\delta) &= A\left|\langle\psi_\pm|\mathrm{e}^{-\mathrm{i}\delta\hat{\Pi}_1}|\psi_{\mathrm{in}}\rangle\right|^2 \\
&= A\left|\langle\psi_\pm|\left[1+\left(\mathrm{e}^{-\mathrm{i}\delta}-1\right)\hat{\Pi}_1\right]|\psi_{\mathrm{in}}\rangle\right|^2 \\
&= A\left|\langle\psi_\pm|\psi_{\mathrm{in}}\rangle\left[1+\left(\mathrm{e}^{-\mathrm{i}\delta}-1\right)w_{\pm,1}\right]\right|^2 \\
&= A\,|\langle\psi_\pm|\psi_{\mathrm{in}}\rangle|^2\left[1+2\left(|w_{\pm,1}|^2-w^{\mathrm{R}}_{\pm,1}\right)(1-\cos\delta)+2\,w^{\mathrm{I}}_{\pm,1}\sin\delta\right] \\
&= A\,|\langle\psi_\pm|\psi_{\mathrm{in}}\rangle|^2\left[1+2\left(|w_{\pm,2}|^2-w^{\mathrm{R}}_{\pm,2}\right)(1-\cos\delta)-2\,w^{\mathrm{I}}_{\pm,2}\sin\delta\right]\,,
\end{aligned}
$$

Amplitude squared — Real part — Imaginary part

Weak values from output intensities

Measured intensity:

$$I_\pm(\delta) = A\,|\langle\psi_\pm|\psi_{\text{in}}\rangle|^2\,\left[1 + 2\left(|w_{\pm,1}|^2 - w^{\mathrm{R}}_{\pm,1}\right)(1-\cos\delta) + 2\,w^{\mathrm{I}}_{\pm,1}\,\sin\delta\right]$$

Weak values from output intensities

Measured intensity:

$$I_{\pm}(\delta) = A\,|\langle\psi_{\pm}|\psi_{\text{in}}\rangle|^2\,\left[1 + 2\left(|w_{\pm,1}|^2 - w^{\text{R}}_{\pm,1}\right)(1 - \cos\delta) + 2\,w^{\text{I}}_{\pm,1}\,\sin\delta\right]$$

Phase

Terms extraction:

$$I_{+}(0) = A\,|\langle\psi_{+}|\psi_{\text{in}}\rangle|^2$$

$$\frac{I_{-}(0) - I_{+}(0)}{4I_{+}(0)} = |w_{+,1}|^2 - w^{\text{R}}_{+,1}$$

$$\frac{I_{+}(\frac{\pi}{2}) - I_{-}(\frac{\pi}{2})}{4I_{+}(0)} = w^{\text{I}}_{+,1}$$

Weak values from output intensities

Measured intensity:

$$I_{\pm}(\delta) = A\,|\langle\psi_{\pm}|\psi_{\text{in}}\rangle|^2\,\left[1 + 2\left(|w_{\pm,1}|^2 - w^{\text{R}}_{\pm,1}\right)(1-\cos\delta) + 2\,w^{\text{I}}_{\pm,1}\,\sin\delta\right]$$

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$$\frac{I_{+}(\frac{\pi}{2}) - I_{-}(\frac{\pi}{2})}{4I_{+}(0)} = w^{\text{I}}_{+,1} \quad ✓$$

Weak values from output intensities

Real part extraction:

$$\frac{I_-(0) - I_+(0)}{4I_+(0)} = |w_{+,1}|^2 - w^{\mathrm{R}}_{+,1} = (w^{\mathrm{R}}_{+,1})^2 + (w^{\mathrm{I}}_{+,1})^2 - w^{\mathrm{R}}_{+,1}$$

Weak values from output intensities

Real part extraction:

$$\frac{I_-(0) - I_+(0)}{4I_+(0)} = |w_{+,1}|^2 - w^{\mathrm{R}}_{+,1} = (w^{\mathrm{R}}_{+,1})^2 + (w^{\mathrm{I}}_{+,1})^2 - w^{\mathrm{R}}_{+,1}$$

Method 1:

$$w^{R}_{+,1} = \frac{1}{2} \pm \frac{1}{2}\sqrt{\frac{I_-(0)}{I_+(0)} - \left(2\, w^{I}_{+,j}\right)^2}$$

✘ Ambiguity in the solution

✓ Reveal nonclassicality

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Method 2:

$$w^{\mathrm{R}}_{+,j} = |w_{+,1}|^2 + \frac{I_+(0) - I_-(0)}{4I_+(0)}$$

✘Extra measurement

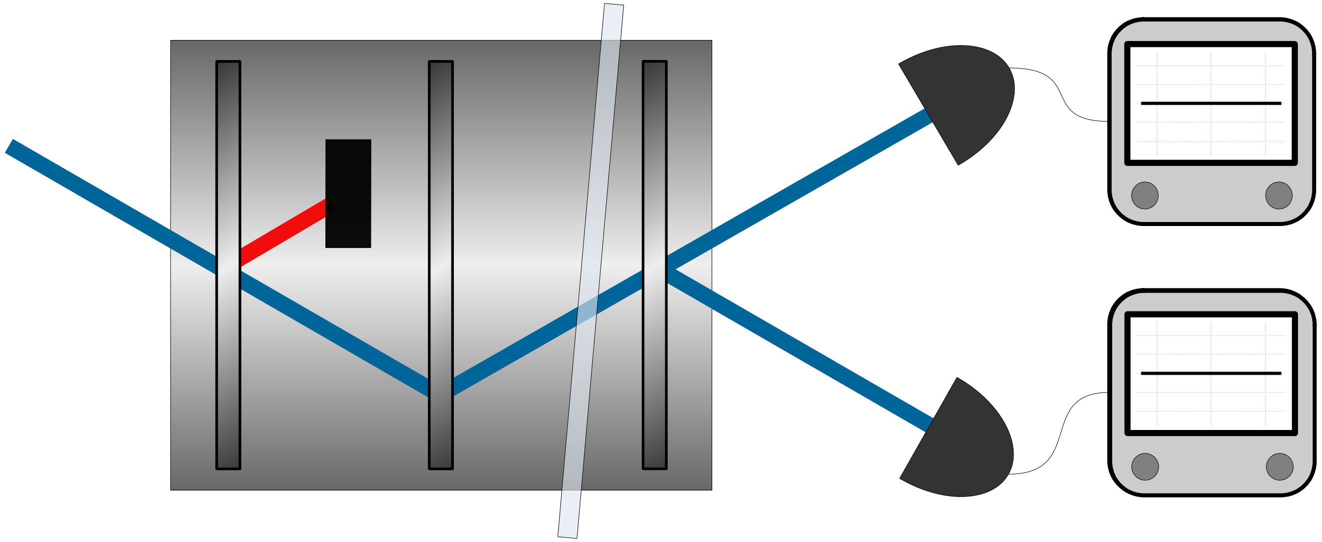
✓Exact solution

Weak values from output intensities

Method 2:

$$w_{+,j}^{\mathrm{R}} = |w_{+,1}|^2 + \frac{I_+(0) - I_-(0)}{4I_+(0)}$$

Path 2 blocked:



Measured intensity:

$$\begin{aligned} I_{+,1} &= A\left|\langle\psi_+|\hat{\Pi}_1|\psi_{\mathrm{in}}\rangle\right|^2 \\ &= A\,|\langle\psi_+|\psi_{\mathrm{in}}\rangle|^2 |w_{+,1}|^2 \\ &= I_+(0)|w_{+,1}|^2 \end{aligned}$$

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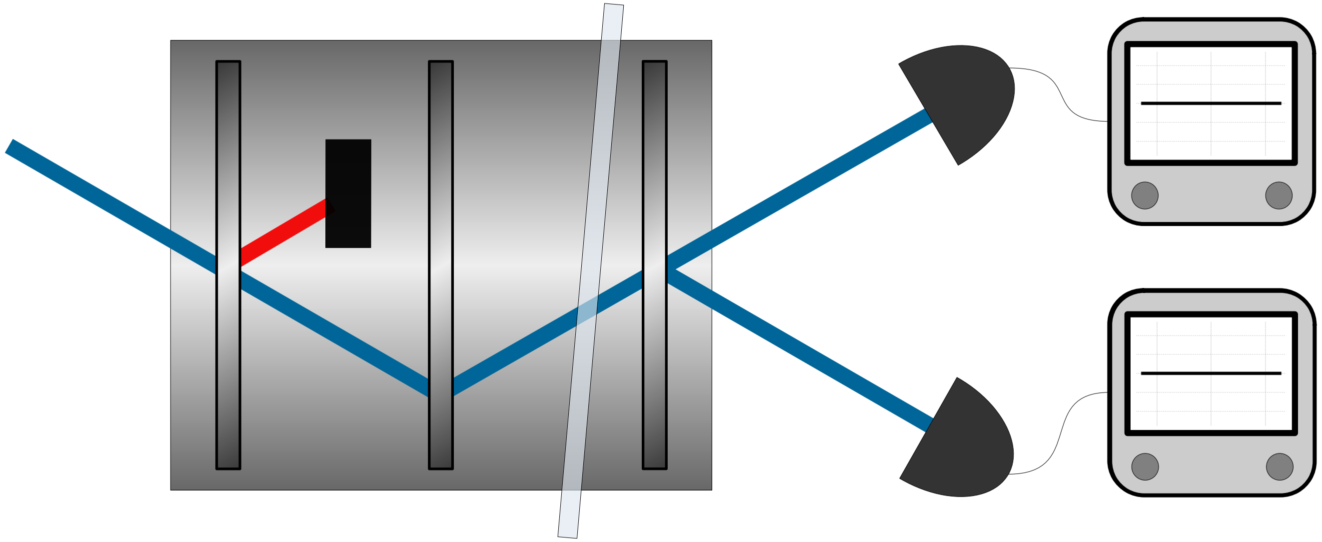
Method 2:

$$w^{\mathrm{R}}_{+,j} = \frac{I_{+,j}}{I_+(0)} + \frac{I_+(0) - I_-(0)}{4I_+(0)}$$

Path 2 blocked:



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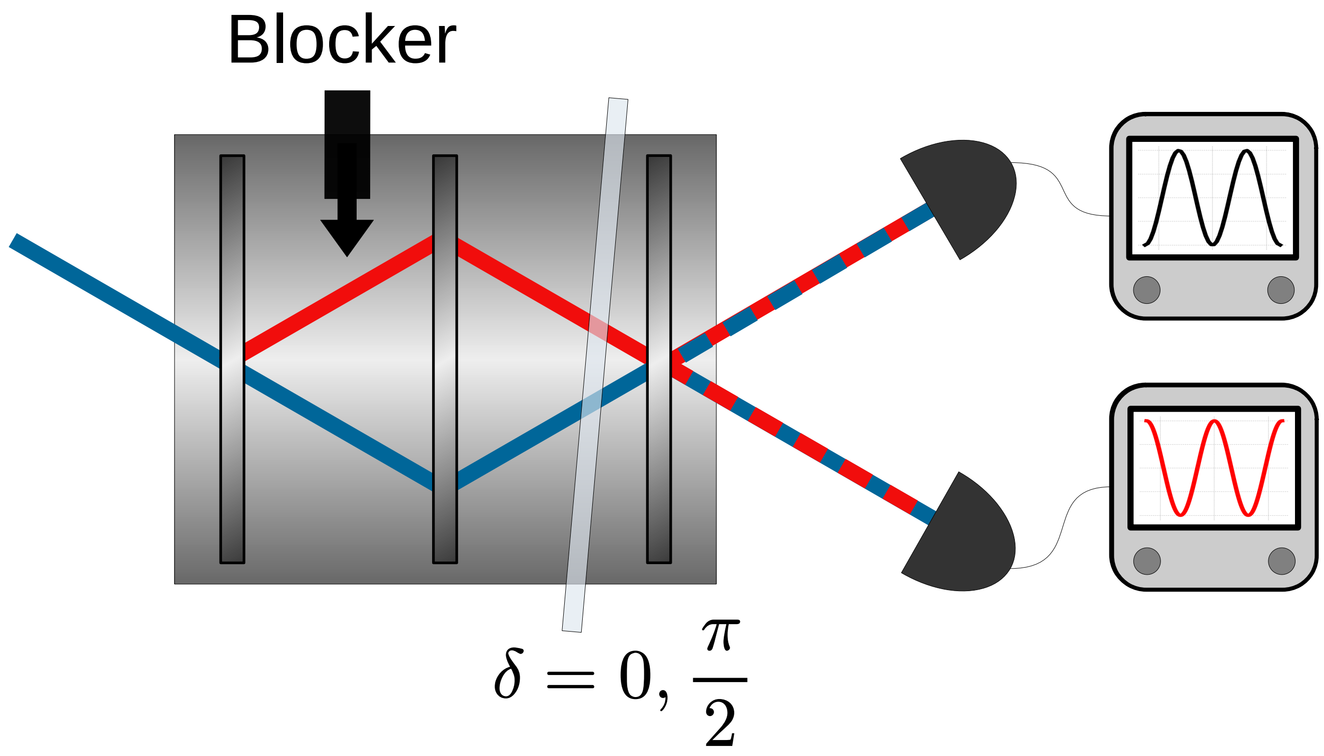

Required setup:

- Interferometer
- Path blocker
- Phase shifter

Measurements:

- No meter state and no weak interaction
- 2 measurements needed to test anomalous features
- 3 measurements needed to completely determine the path weak-values

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